Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION FOR STAIRCASE FOUNDATION -49W1 Raw Water Treatment Plant

Document Number:	**MD-216-49W1-EG-CV-CAL-1005**
Revision:	**D01**
Revision Date:	**08.03.2024**
Status:	**AS-BUILT**

Outotec Document Number:	900707	GAU01	AG11	PED05	49005	OU100732593	R3
L&T Document Number:	O19045	080	C	K	24890	005	R3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	29.10.2021	Issued for approval	NanWu	MauNot	LuiPiz	
A02	23.11.2021	Issued for approval	NanWu	MauNot	LuiPiz	
C01	14.02.2022	Issued for construction	NanWu	MauNot	LuiPiz	
D01	08.03.2024	AS-BUILT - Cover updated, Content UnChanged	tadcla	---	tadcla	

TABLE OF CONTENTS

1. Intent 3
2. Scope 3
3. Material property 3
4. Load calculations 4
   4.1. Design loads 4
   4.2. Load combinations 4
5. Soil bearing capacity 5
6. Calculation analysis 6
   6.1. Calculation model in RFEM 6
   6.2. Soil bearing check 15
   6.3. Overturning check 21
   6.4. Sliding check 25
   6.5. Design of footing column 29
   6.6. Design of pad footing 38

References 52

Attachment 1 Input load data from staircase 53

Attachment 2 Comment resolution sheet 54

PROPRIETARY & CONFIDENTIAL

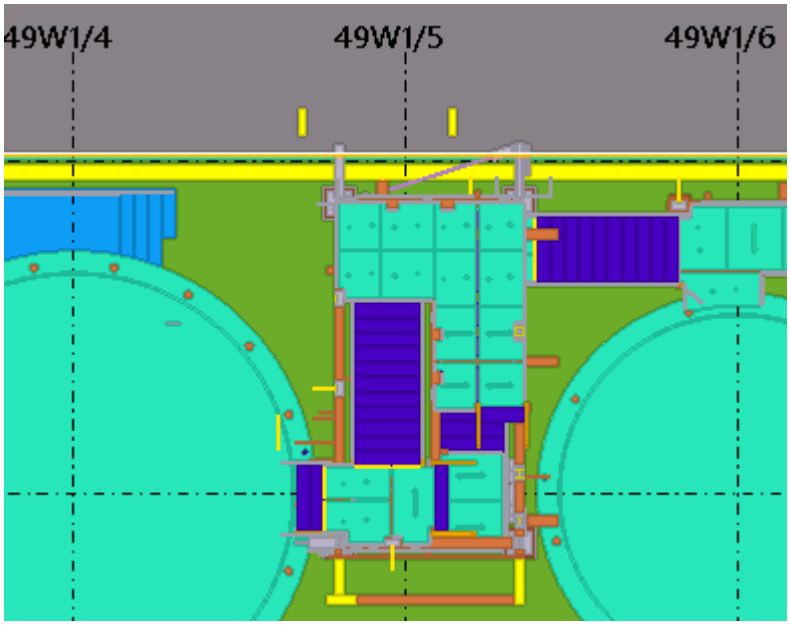
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This calculation report is to design and analyze staircase foundations in Raw Water Treatment Plant area. The report describes the methodology and the design parameters used in the design calculation.

2. SCOPE



The design follows and verified with requirements in Saudi Building Code:

- SBC_301 (Structural – Loading and Forces)
- SBC_304 (Structural – Concrete Structures)
- SBC_303 (Structural – Soil and Foundations)

Concrete structures are modeled as surface and member in Dlubal RFEM 5.21, both strength and allowable stress calculations are included.

3. MATERIAL PROPERTY

Concrete compressive strength at 28 days: f_{ck} = 35 N/mm^2

Reinforcing bars ASTM A615- Grade 60, yield strength f_{yk} = 420 N/mm^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. LOAD CALCULATIONS

4.1. Design loads

All design loads are results from staircase in MD-216-49W1-EG-ST-CAL-1001. Input load values are given in Attachment 1. Basic load cases include:

1	Self-weight of foundation and back soil weight (D)
2	Live load (L)
3	Wind load in ±X and ±Y direction (W)
4	Temperature ±25 (T)
5	Seismic load (E)

4.2. Load combinations

For strength calculation:

Load Combination	Expression
101	1,40 D
102	1,40 (D +T) + 1,70 L
103	1,20 D + 1,00 L
104	1,20 D + 0,80 W
105	1,20 D + 1,60 W + 1.00 L
106	1,20 D + 1,00 E + 1,00 L
107	0,90D + 1,60 W
108	0,90 E + 1,00 E

For allowable stress calculation:

Load Combination	Expression
201	D
202	D + L + T
203	D + W
204	D + 0,70 E
205	D + 0,75 W + 0,75 L

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

206	D+ 0,75 (0,70 E) + 0,75 L
207	0,60 D + W
208	0,60 D + 0,70 E

5. SOIL BEARING CAPACITY

Vertical modulus of subgrade reaction $K_s = 120q_a = 33750$ kN/m^3 at 0.6m below FGL is given in geotechnical report MD-216-3000-EG-CV-RPT-1008-C01, BH08 and applied in RFEM as spring constant in z direction. Horizontal modulus of subgrade reaction $K_s = 155{,}000$ kN/m^2 is applied as line support in x and y directions.

Recommended allowable bearing capacity $\sigma_{s,max} = 375$ kN/m^2 is given in geotechnical report. Therefore, the contact stress results shall not exceed the limit.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. CALCULATION ANALYSIS

6.1. Calculation model in RFEM

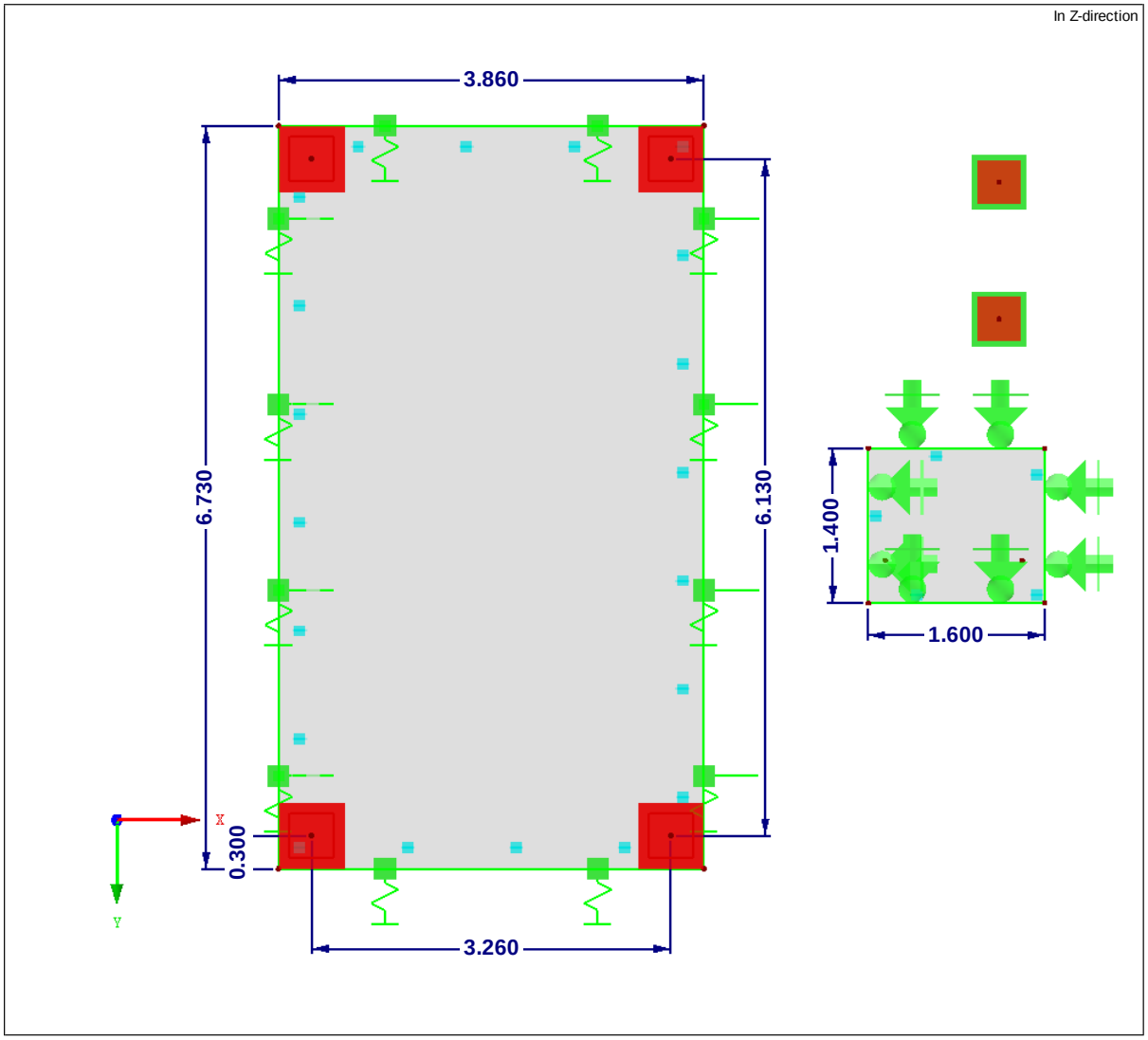


Figure 1 Calculation model in RFEM

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

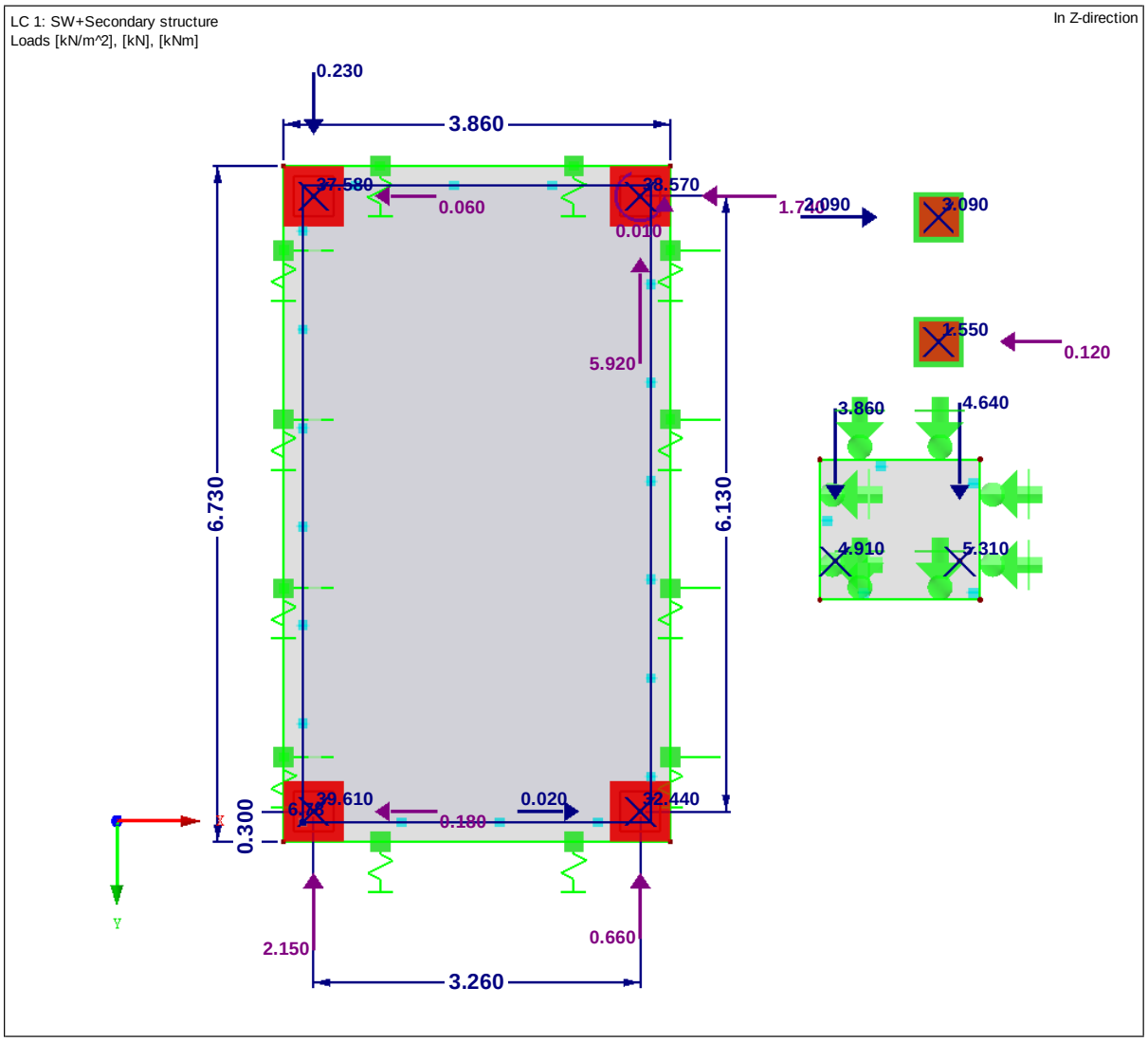


Figure 2 Dead load

dead load includes self-weight and backfill soil

backfill soil weight (6.73m*3.86m - 4*0.6m*0.6m) * 18kN/m^3 * 0.4m = 176.67kN

apply as surface load 176.67kN/ (6.73m*3.86m) = 6.80kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

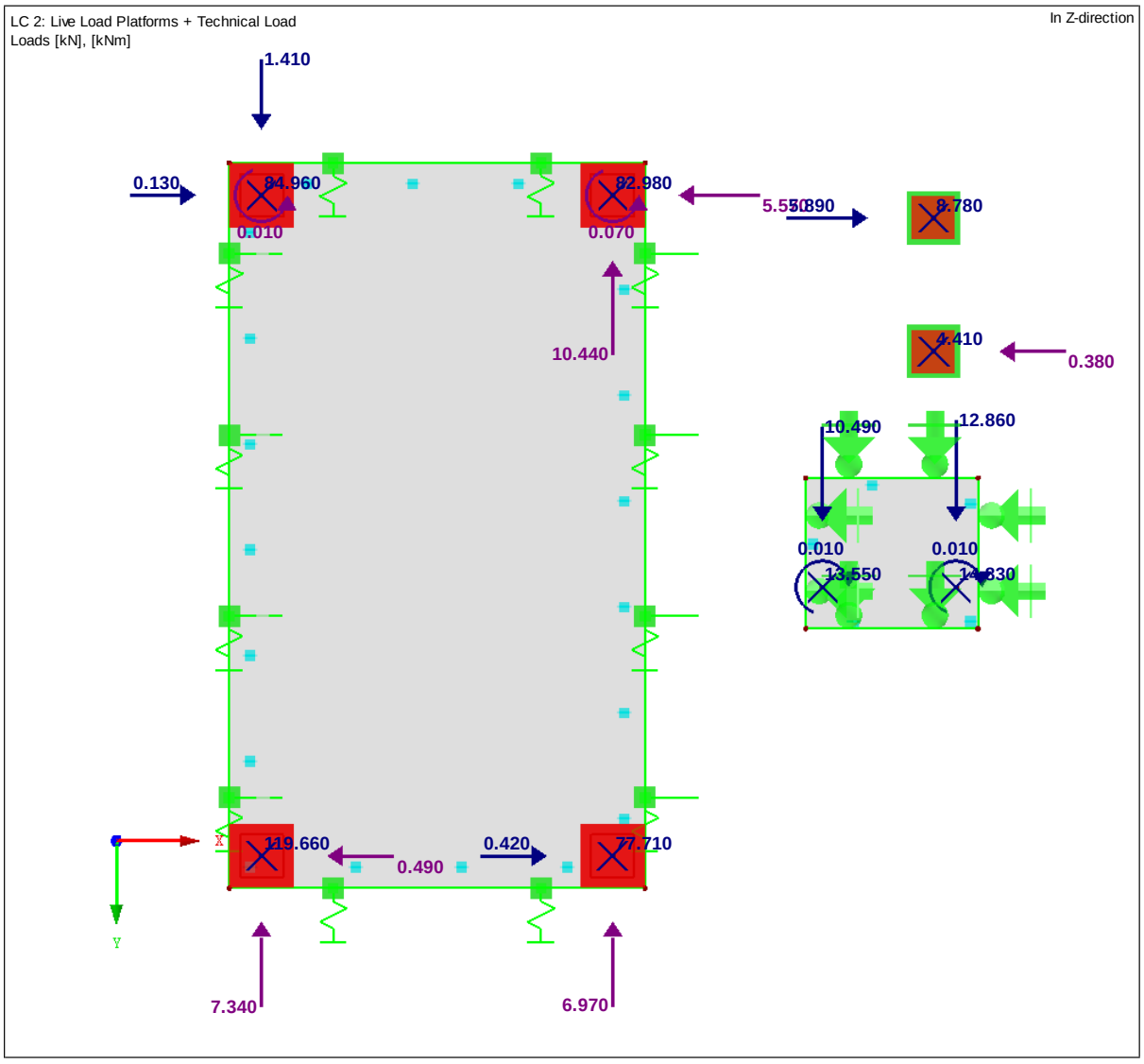


Figure 3 Live load

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

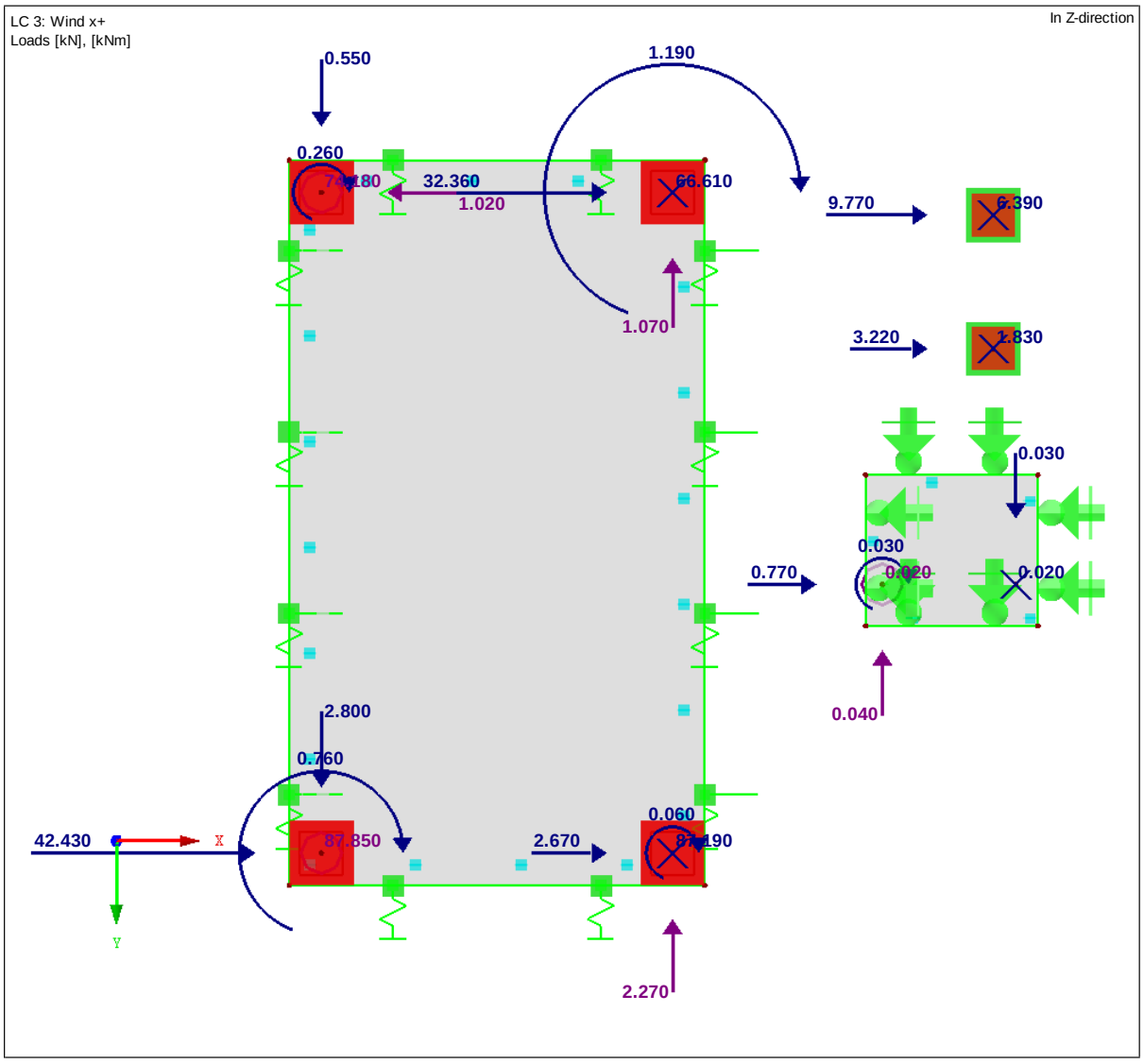


Figure 4 Wind load in X+

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

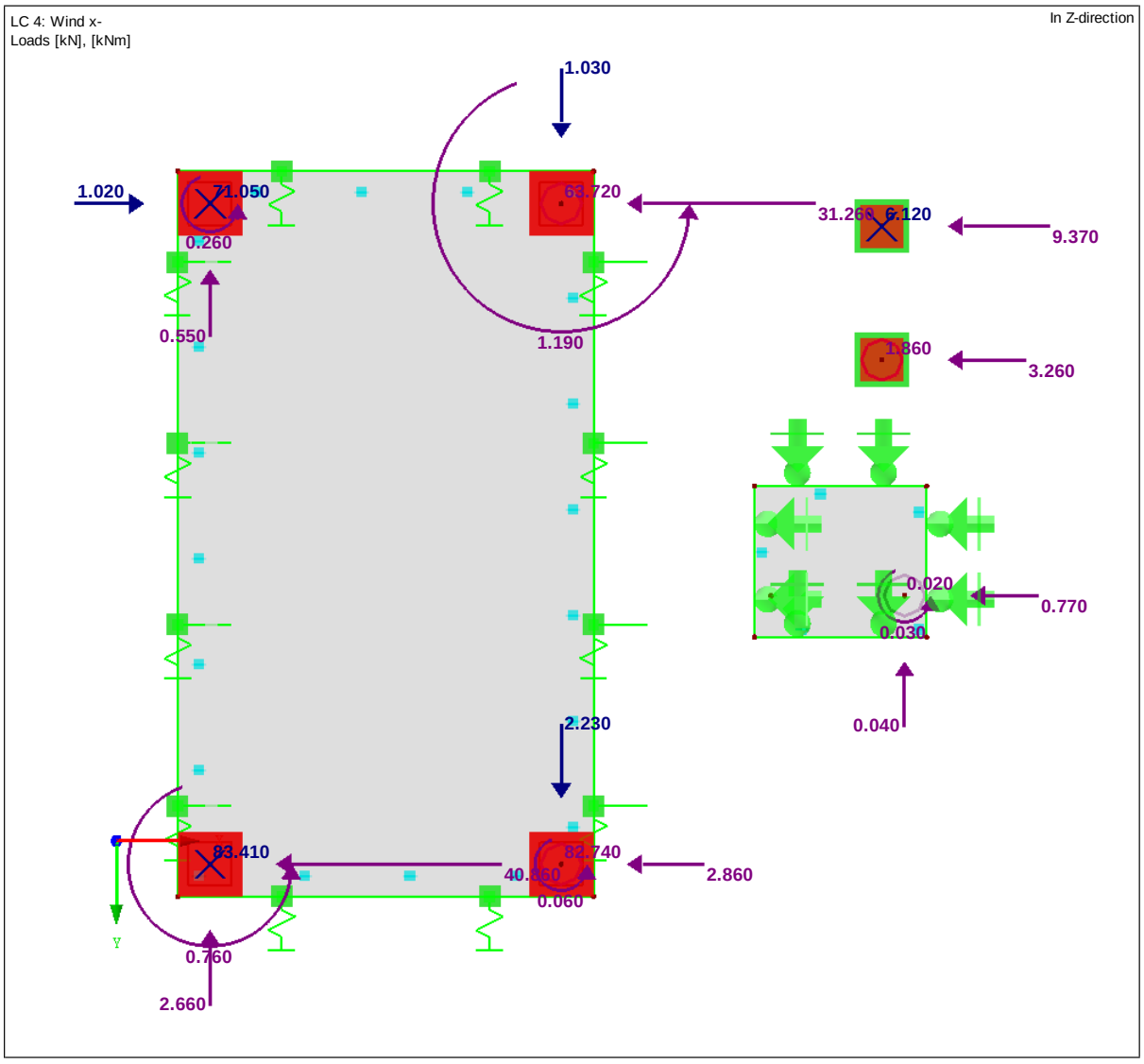


Figure 5 Wind load in X-

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

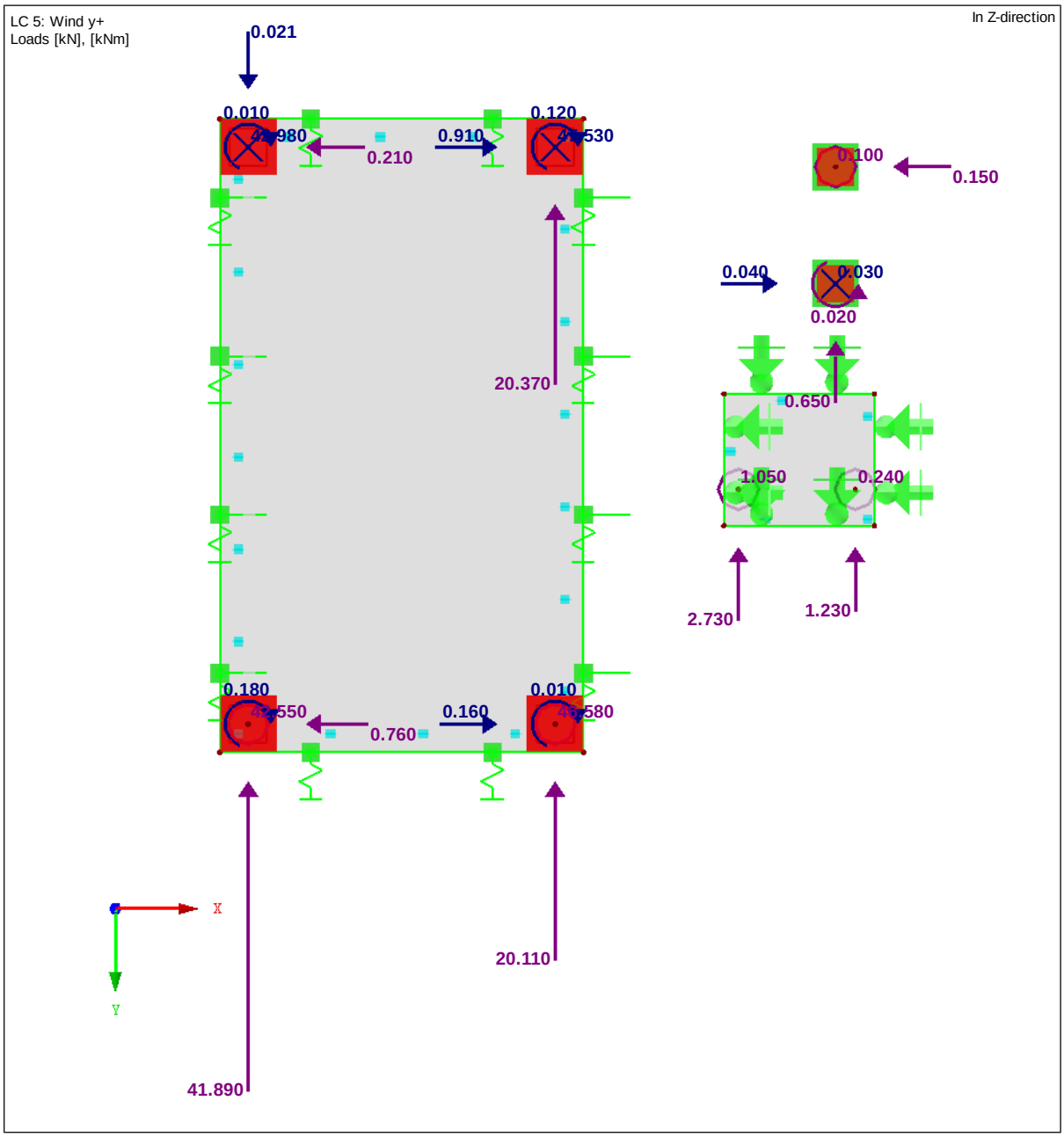


Figure 6 Wind load in Y+

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

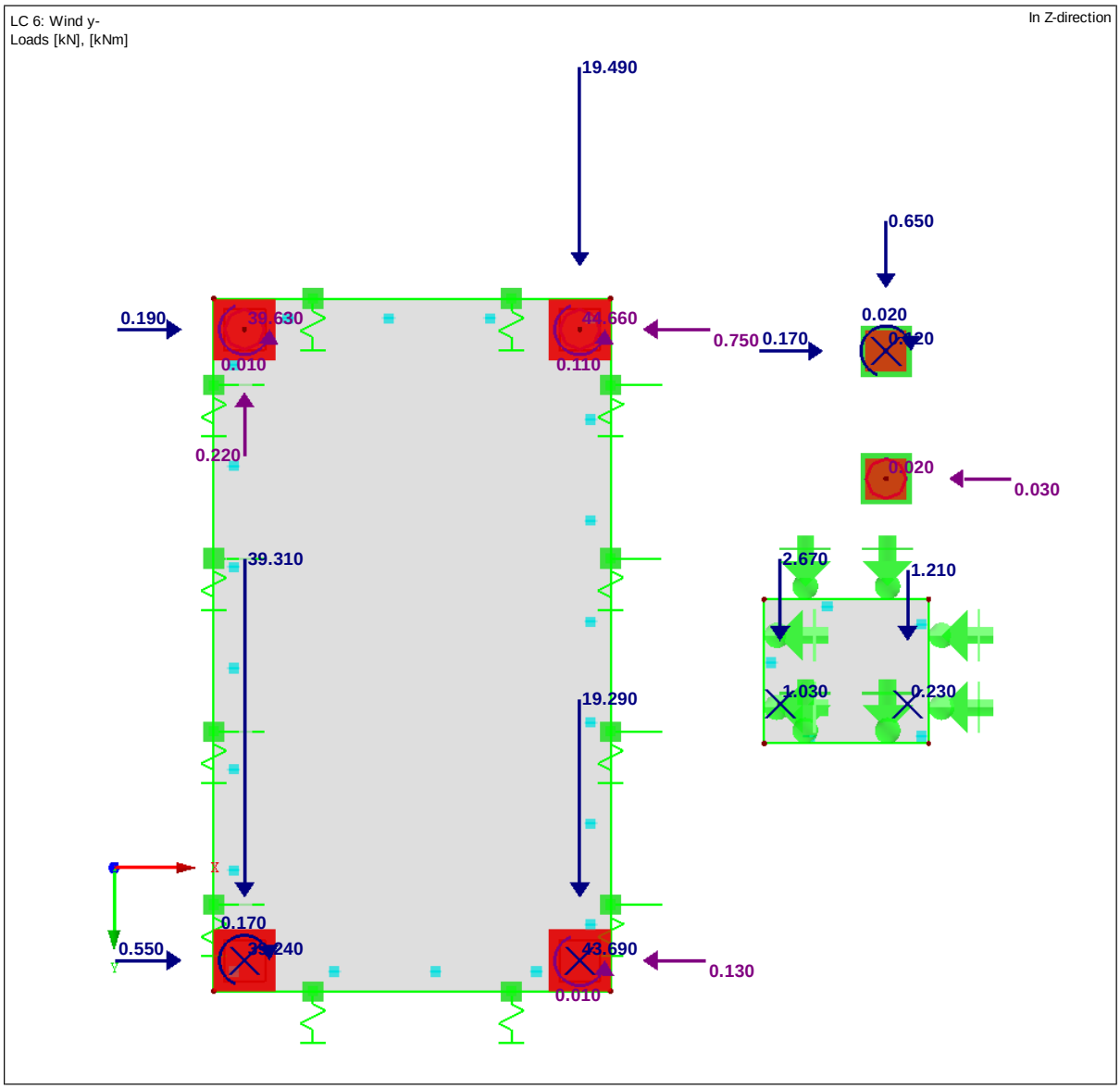


Figure 7 Wind load in Y-

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

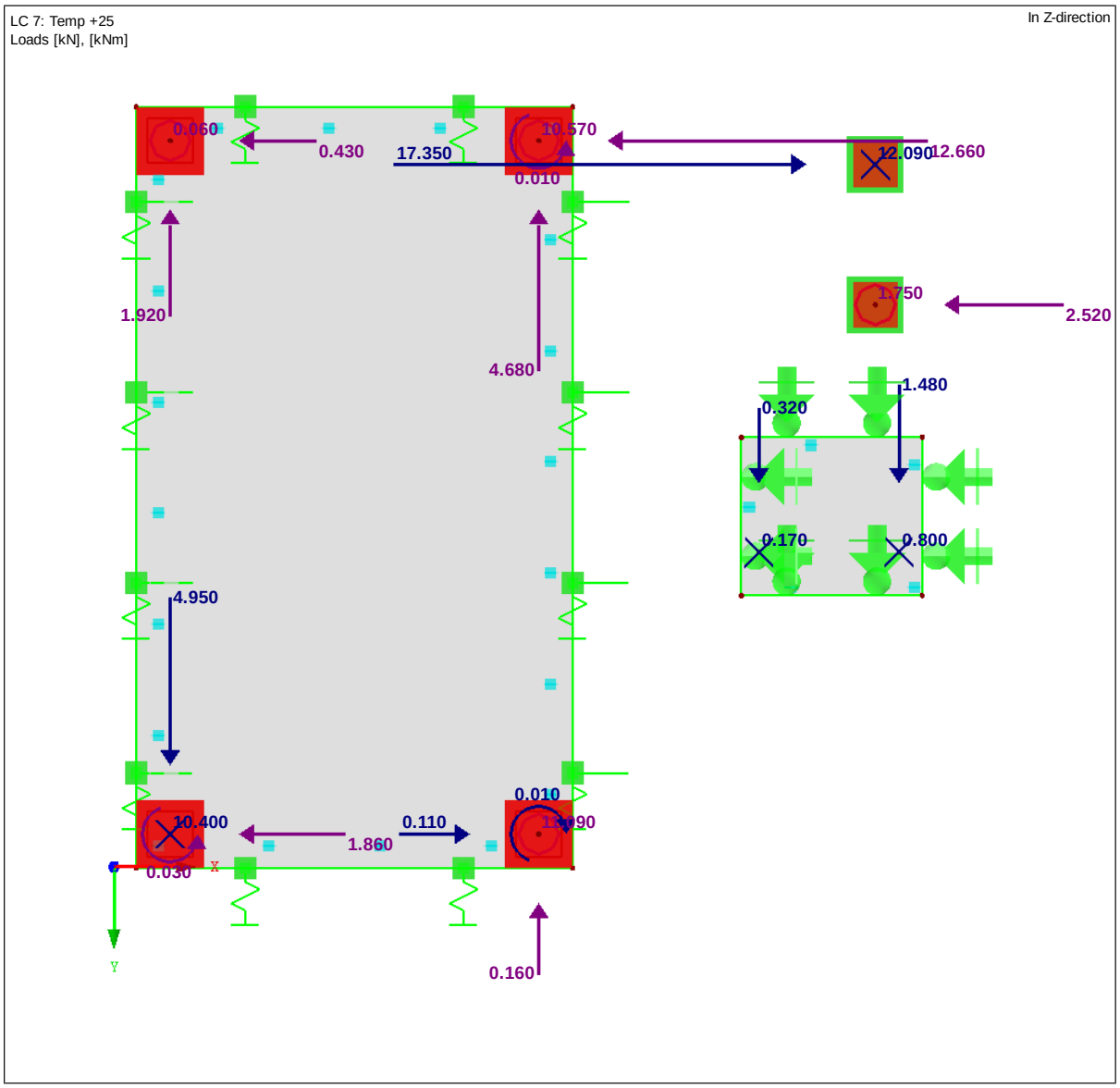


Figure 8 Temperature load +25

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

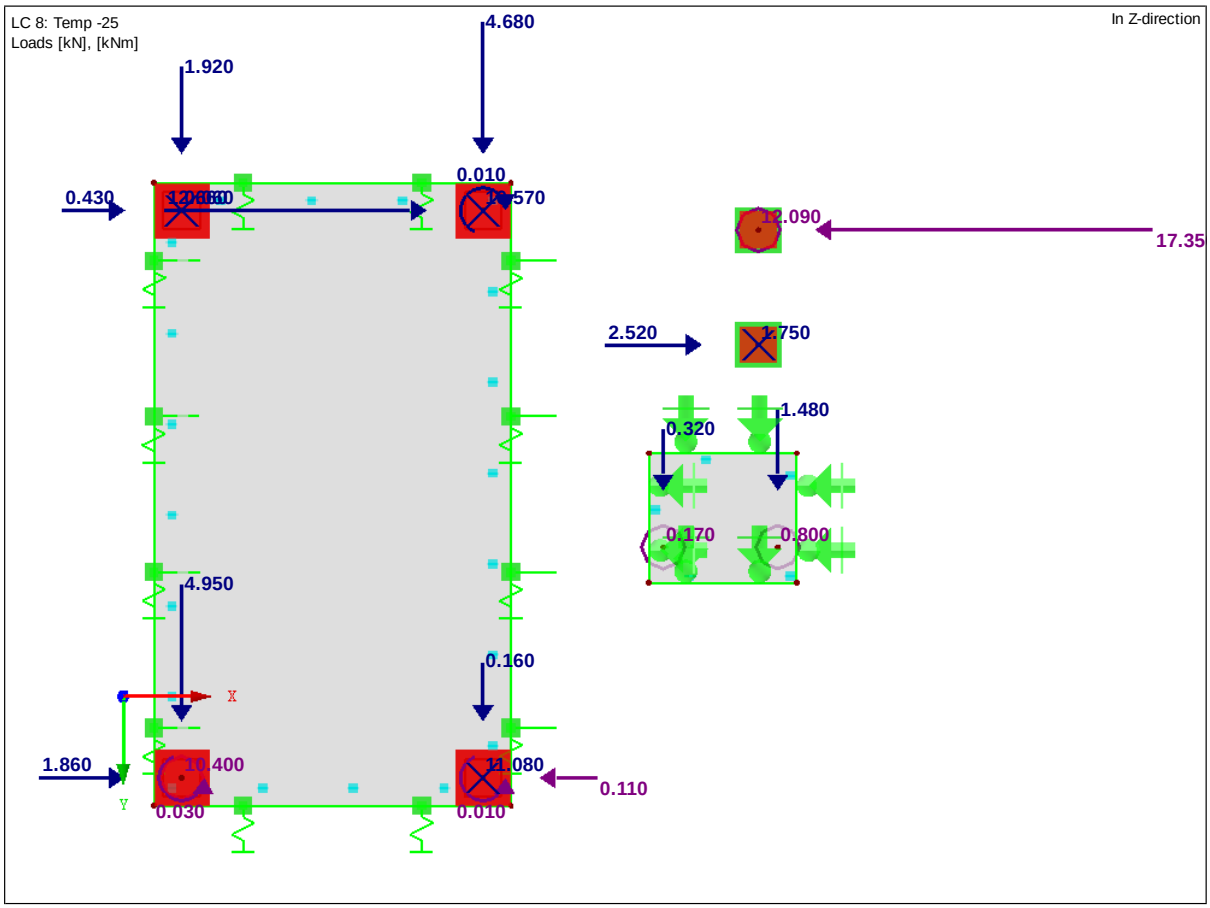


Figure 9 Temperature load -25

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6.2. Soil bearing check

soil pressure σ_z is shown as contact pressure of the surface in RFEM.

σ_z = Ned/A ± Mx/Wx ± My/Wy

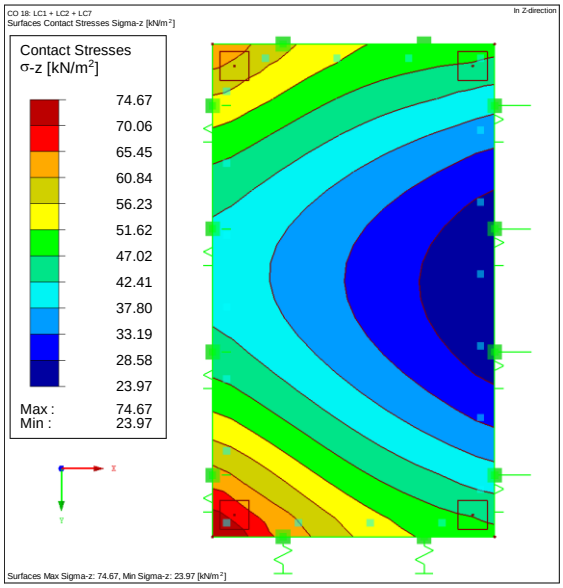


Figure 10 Contact stress for load combination 17

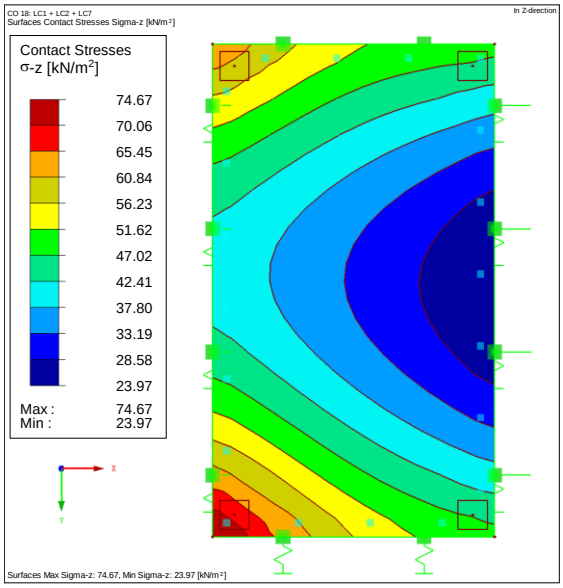


Figure 11 Contact stress for load combination 18

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

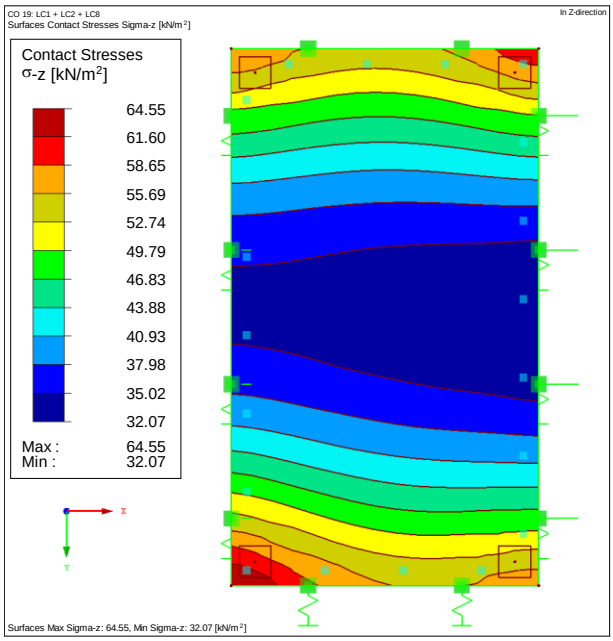


Figure 12 Contact stress for load combination 19

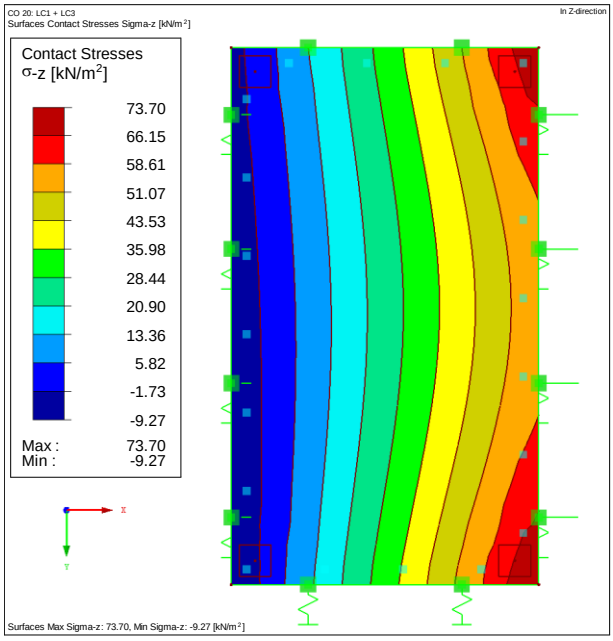


Figure 13 Contact stress for load combination 20

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

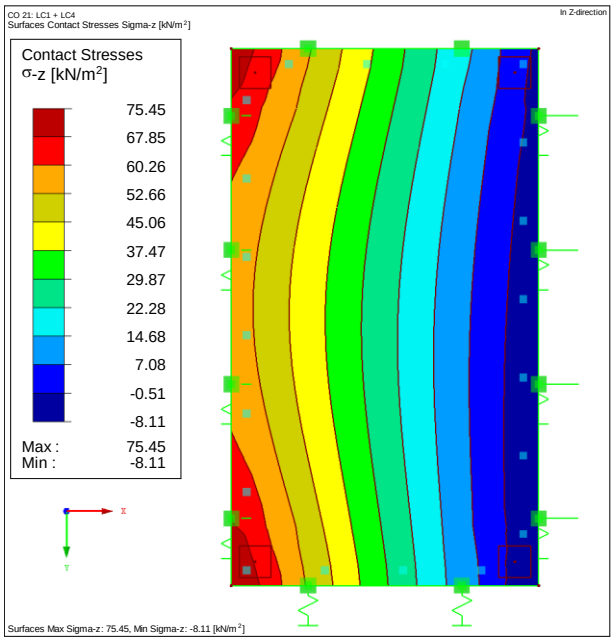


Figure 14 Contact stress for load combination 21

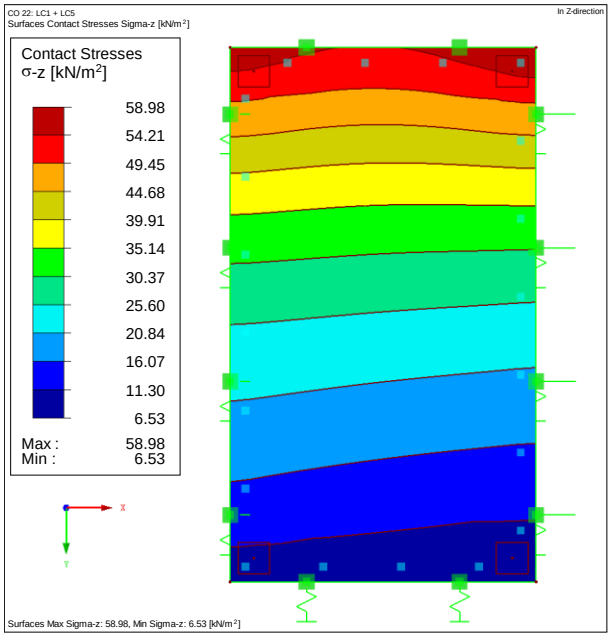


Figure 15 Contact stress for load combination 22

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

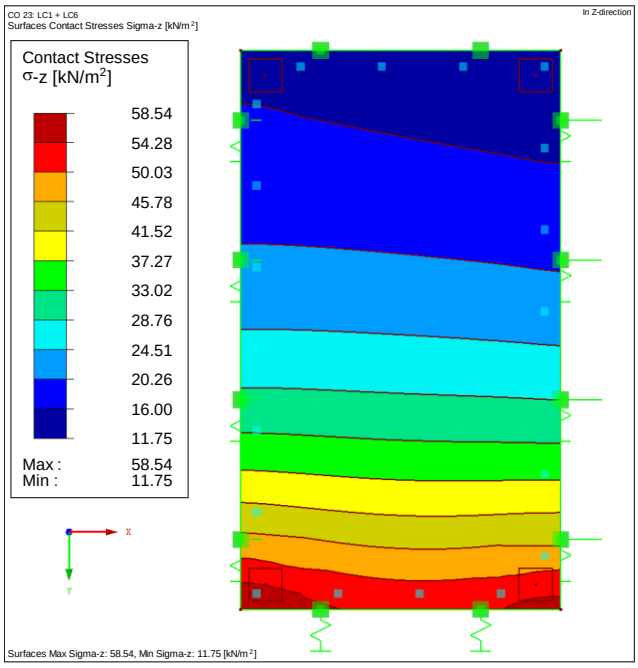


Figure 16 Contact stress for load combination 23

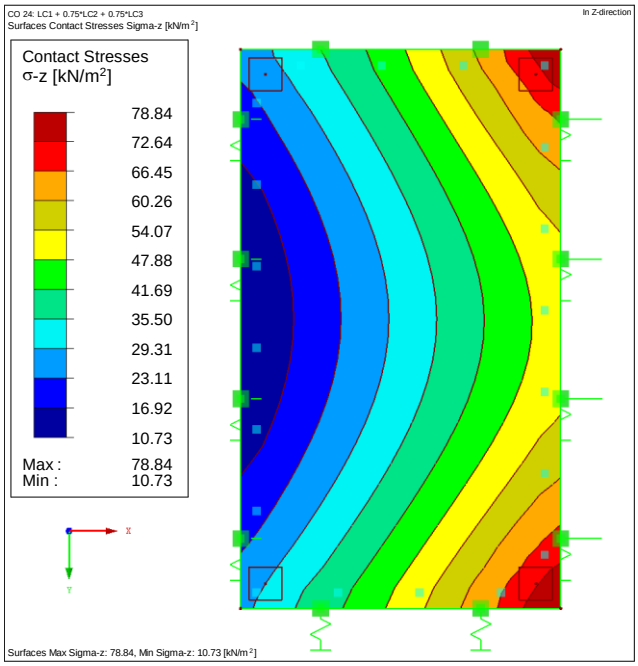


Figure 17 Contact stress for load combination 24

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

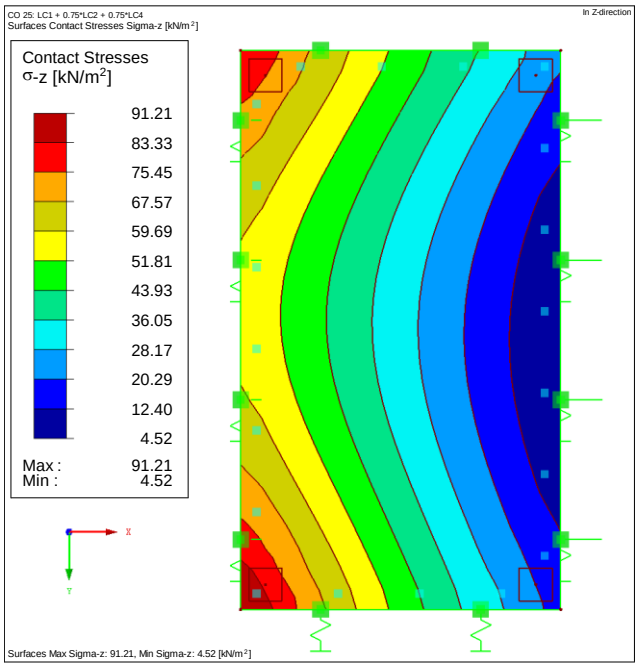


Figure 18 Contact stress for load combination 25

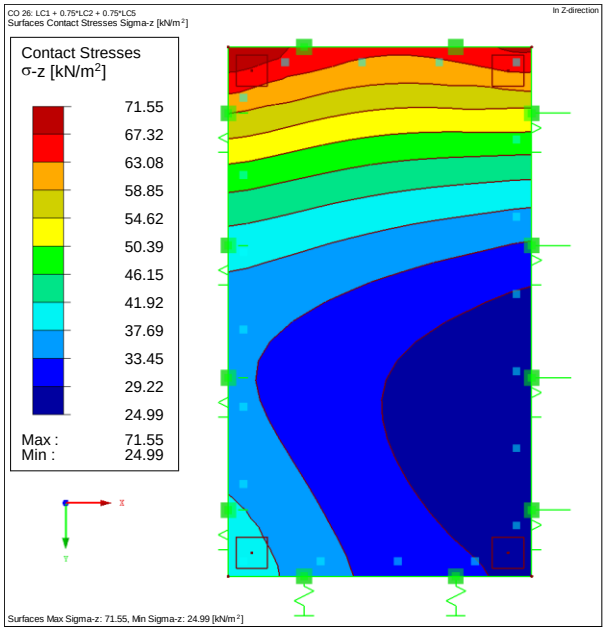


Figure 19 Contact stress for load combination 26

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

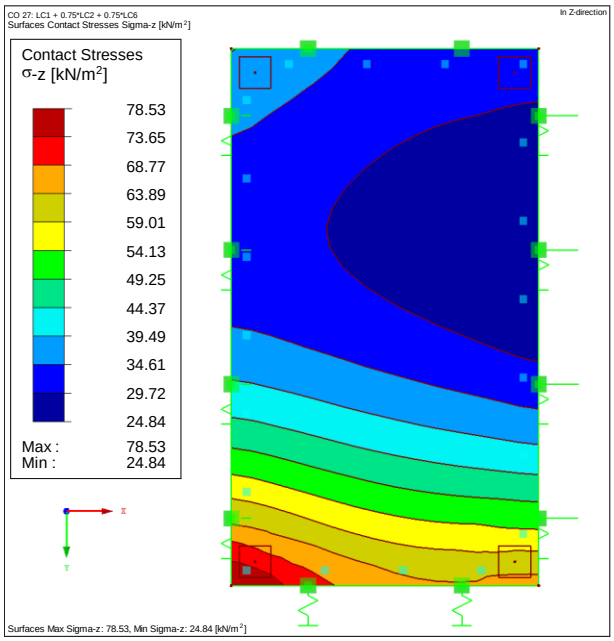


Figure 20 Contact stress for load combination 27

C01

$\sigma_s < \sigma_{s,\,max} = 375 kN/m^2$, ok

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6.3. Overturning check

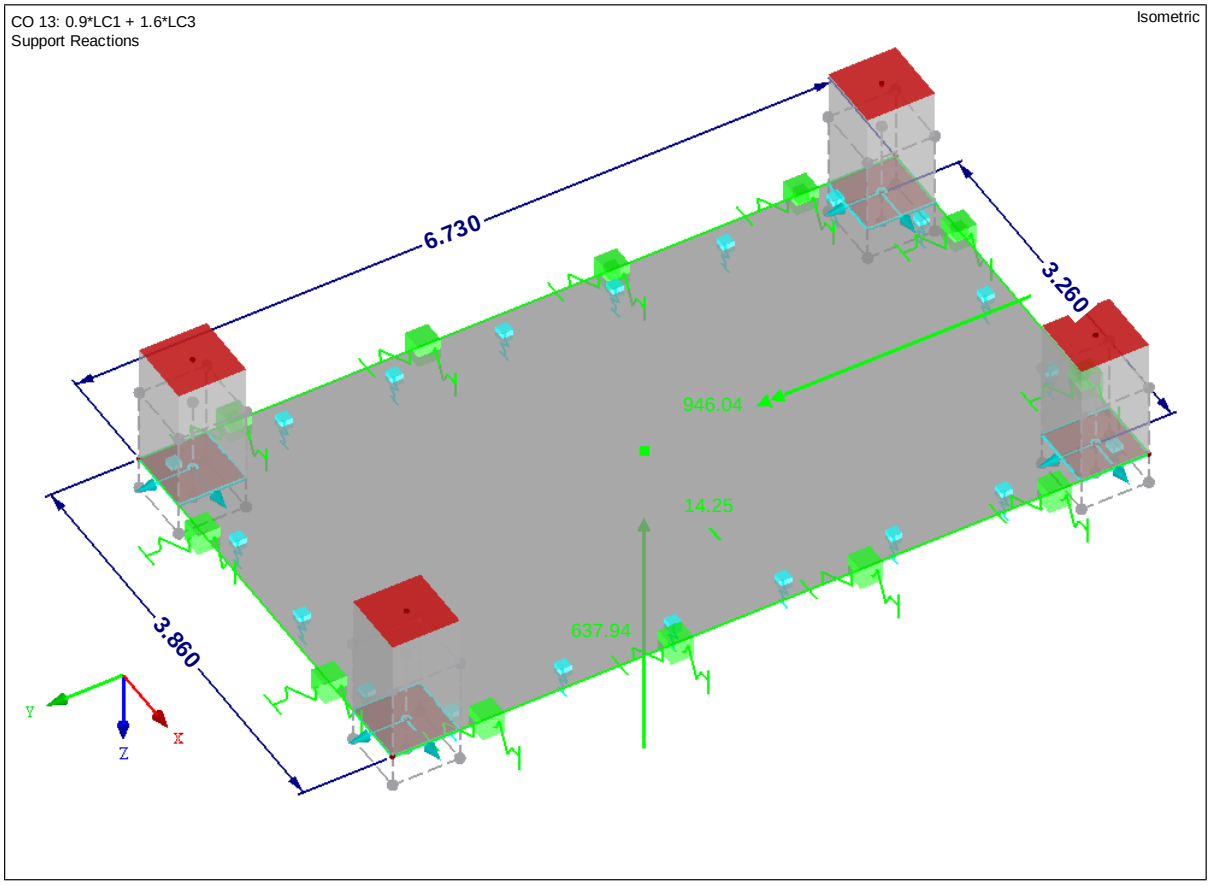


Figure 21 Resultant support reactions for load combination 13

check overturning at load case 0.90D+1.60W,
eccentricity e = My / Ned = 946.04kNm / 637.94kN = 1.48m
B/2 = 3.86m / 2 = 1.93m
e / (B/2) = 0.77 < 1.0, overturning ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

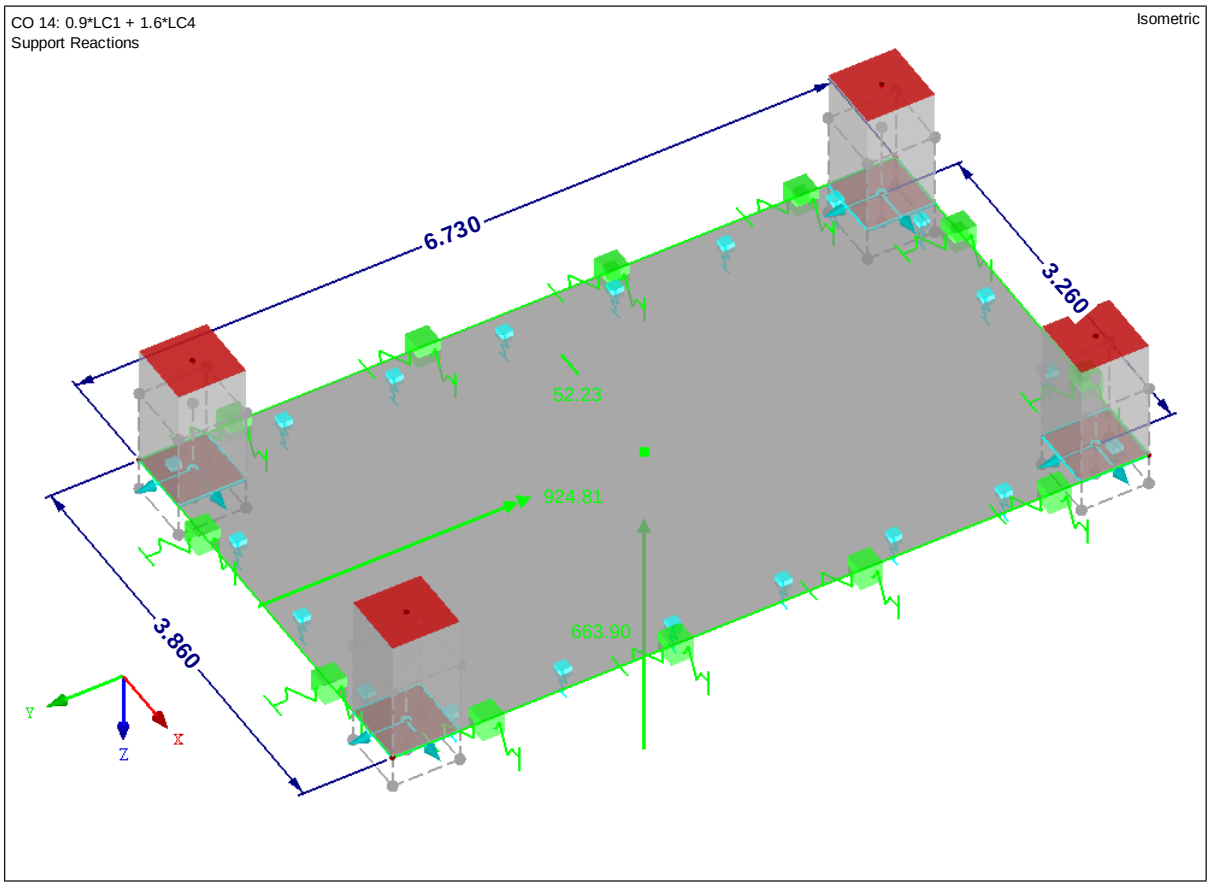


Figure 22 Resultant support reactions for load combination 14

check overturning at load case 0.90D+1.60W,
eccentricity e = My / Ned = 924.81kNm / 663.90kN = 1.39 m
B/2 = 3.86m / 2 = 1.93m
e / (B/2) = 0.72 < 1.0, overturning ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

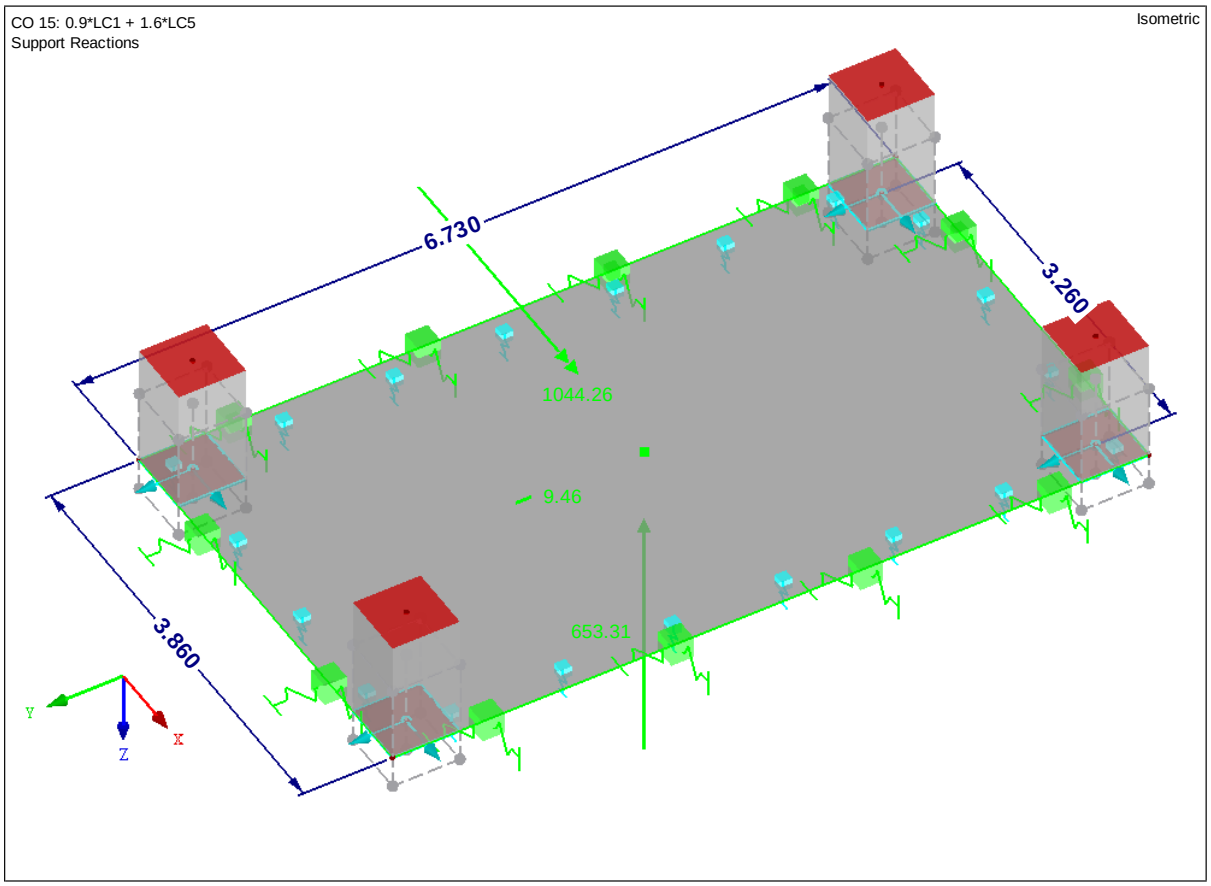


Figure 23 Resultant support reactions for load combination 15

check overturning at load case 0.90D+1.60W,
eccentricity e = My / Ned = 1044.26kNm / 653.31kN = 1.53 m
B/2 = 6.73m / 2 = 3.365m
e / (B/2) = 0.48 < 1.0, overturning ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

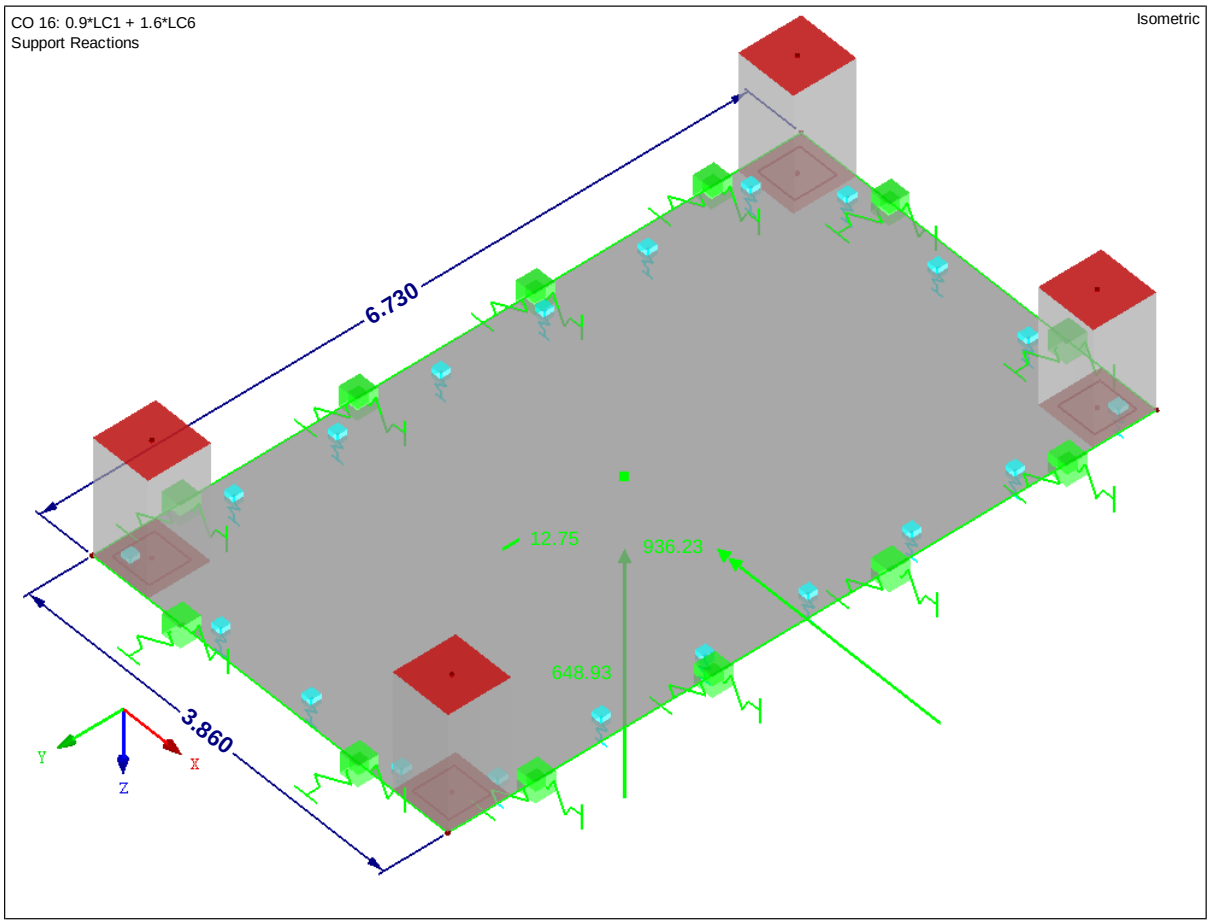


Figure 24 Resultant support reactions for load combination 16

check overturning at load case 0.90D+1.60W,
eccentricity e = My / Ned = 936.23kNm / 648.93kN = 1.44 m
B/2 = 6.73m / 2 = 3.365m
e / (B/2) = 0.43 < 1.0, overturning ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6.4. Sliding check

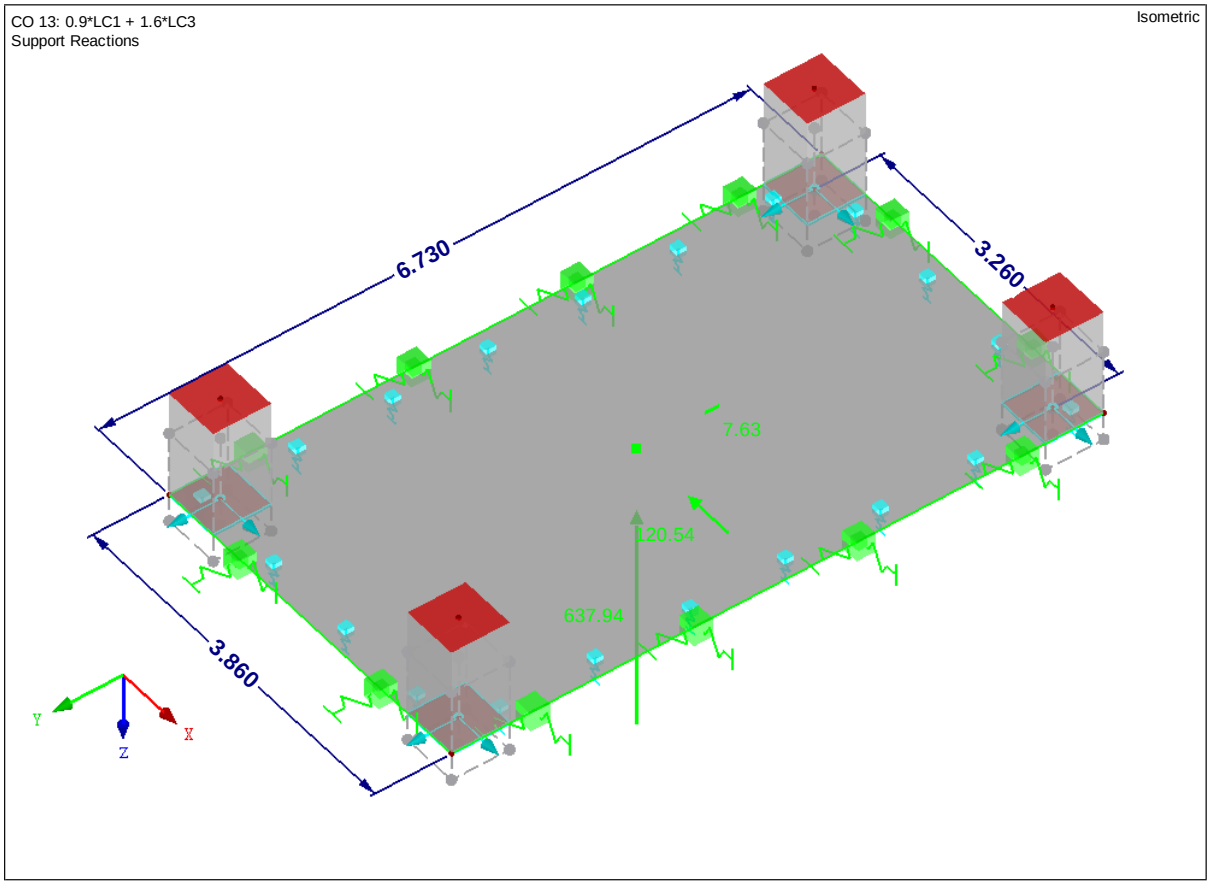


Figure 25 Resultant support reactions for load combination 13

applied horizontal forces

$Hd = [(Hx)^2 + (Hy)^2]^{1/2} = 120.78kN$

sliding resistance

$Rd = 0.50 * Ned = 318.97kN$

sliding check $1.50 Hd/ (0.90 Rd) = 0.63 < 1.0$, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

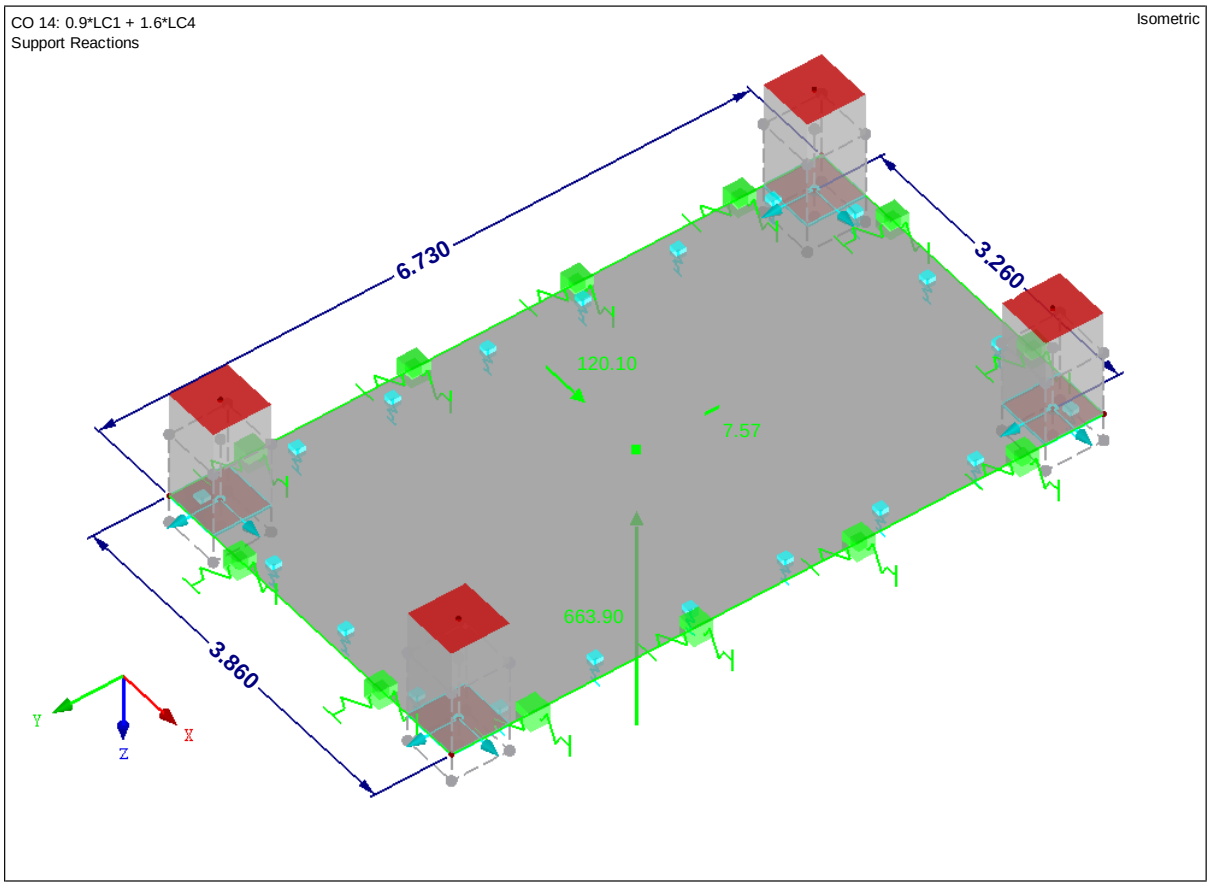


Figure 26 Resultant support reactions for load combination 14

applied horizontal forces

$Hd = [(Hx)^2 + (Hy)^2]^{1/2} = 120.34kN$

sliding resistance

$Rd = 0.50 * Ned = 318.95kN$

sliding check $1.50 Hd/ (0.90Rd) = 0.64 < 1.0$, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

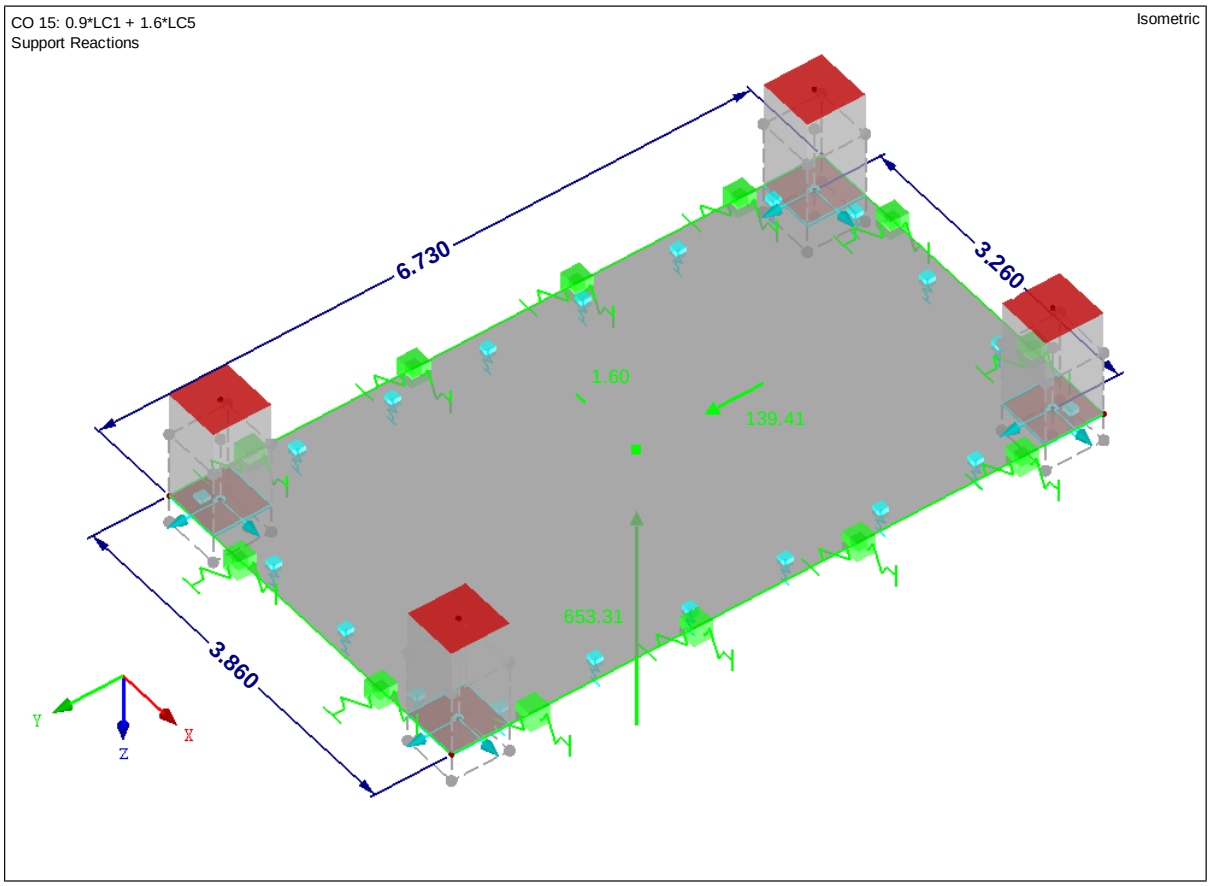


Figure 27 Resultant support reactions for load combination 15

applied horizontal forces

$Hd = [(Hx)^2 + (Hy)^2]^{1/2} = 139.42kN$

sliding resistance

$Rd = 0.50 * Ned = 326.655kN$

sliding check $1.50 Hd/ (0.90Rd) = 0.62 < 1.0$, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

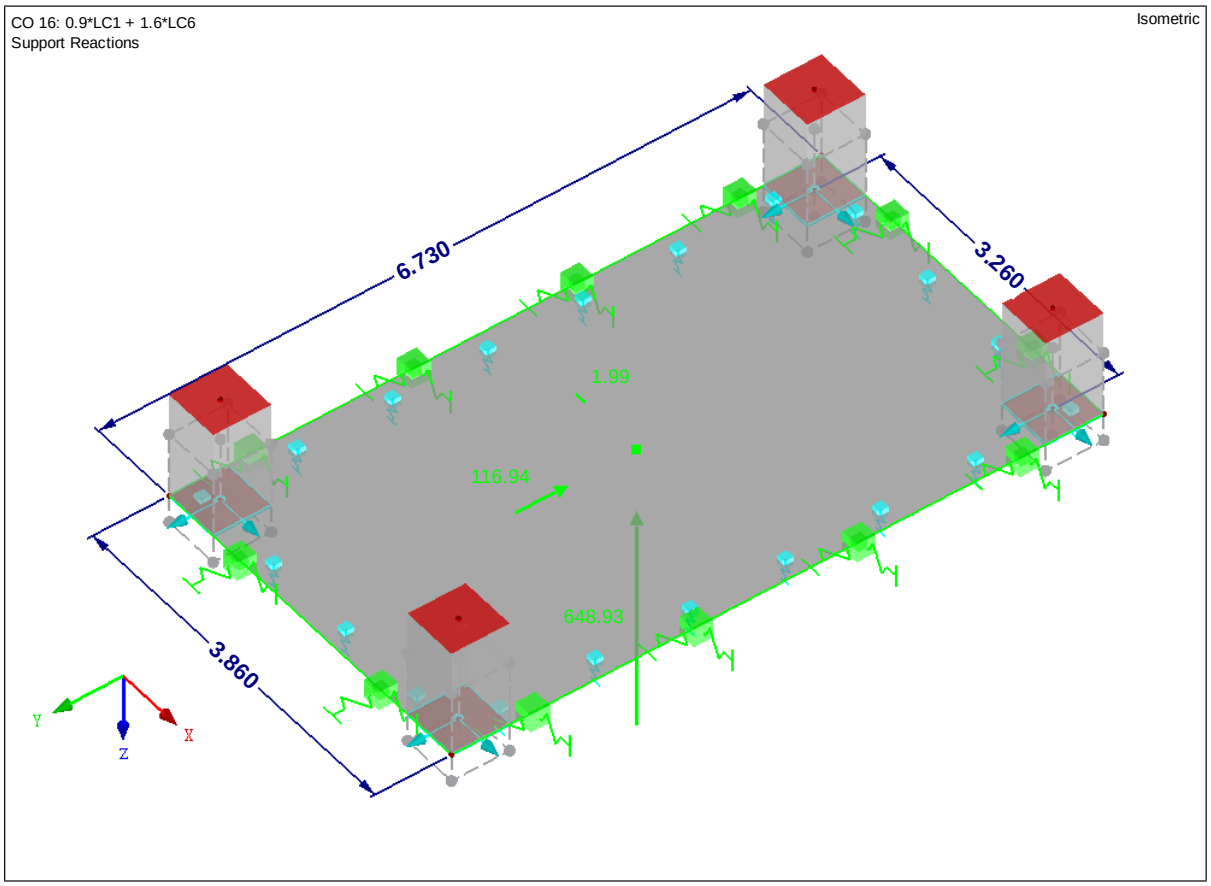


Figure 28 Resultant support reactions for load combination 16

applied horizontal forces

$Hd = [(Hx)^2 + (Hy)^2]^{1/2} = 116.96kN$

sliding resistance

$Rd = 0.50 * Ned = 324.465kN$

sliding check $1.50 Hd/ (0.90Rd) = 0.73 < 1.0$, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

C01

6.5. Design of footing pedestal

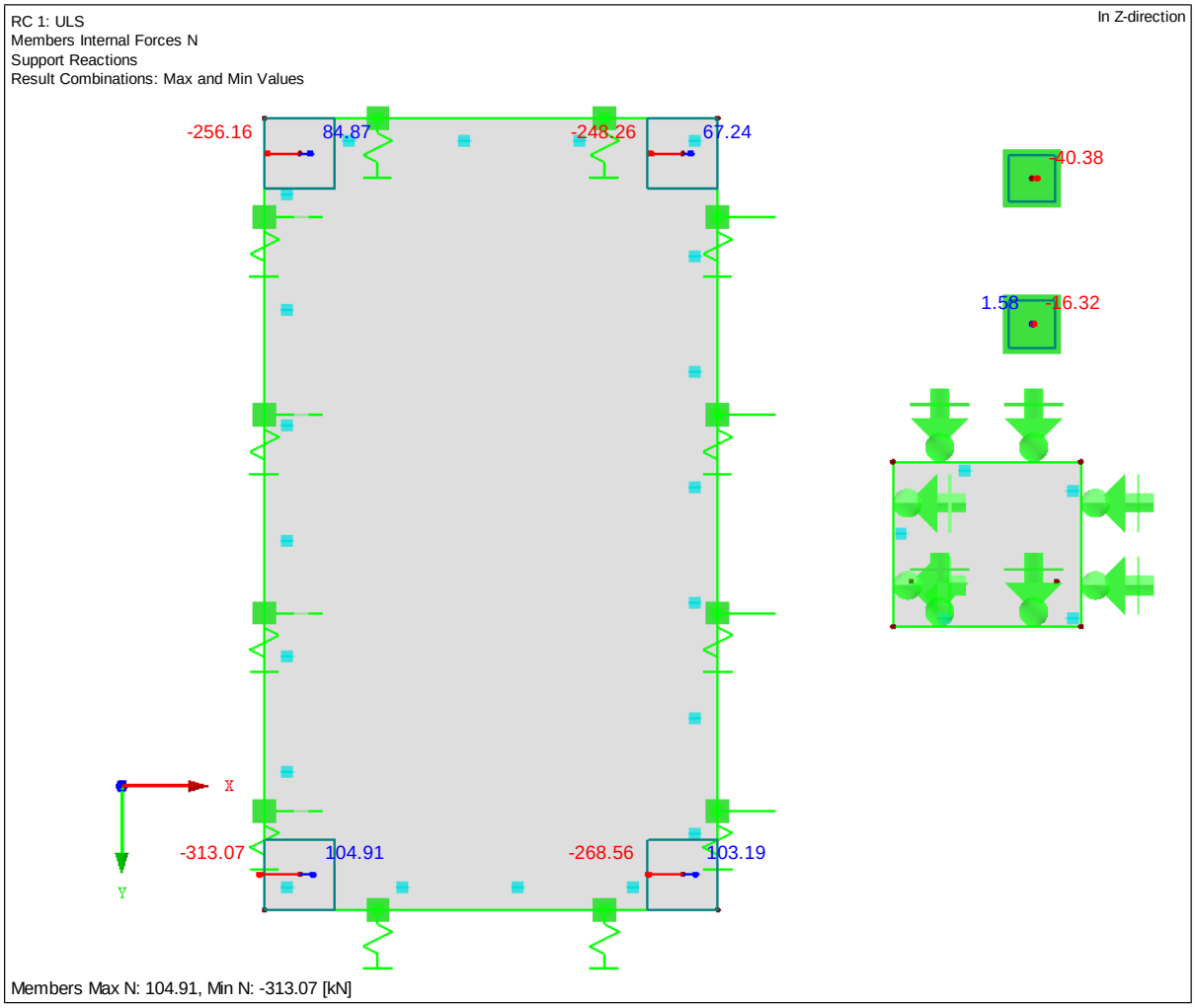


Figure 29 Axial force $N_{Ed,max}$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

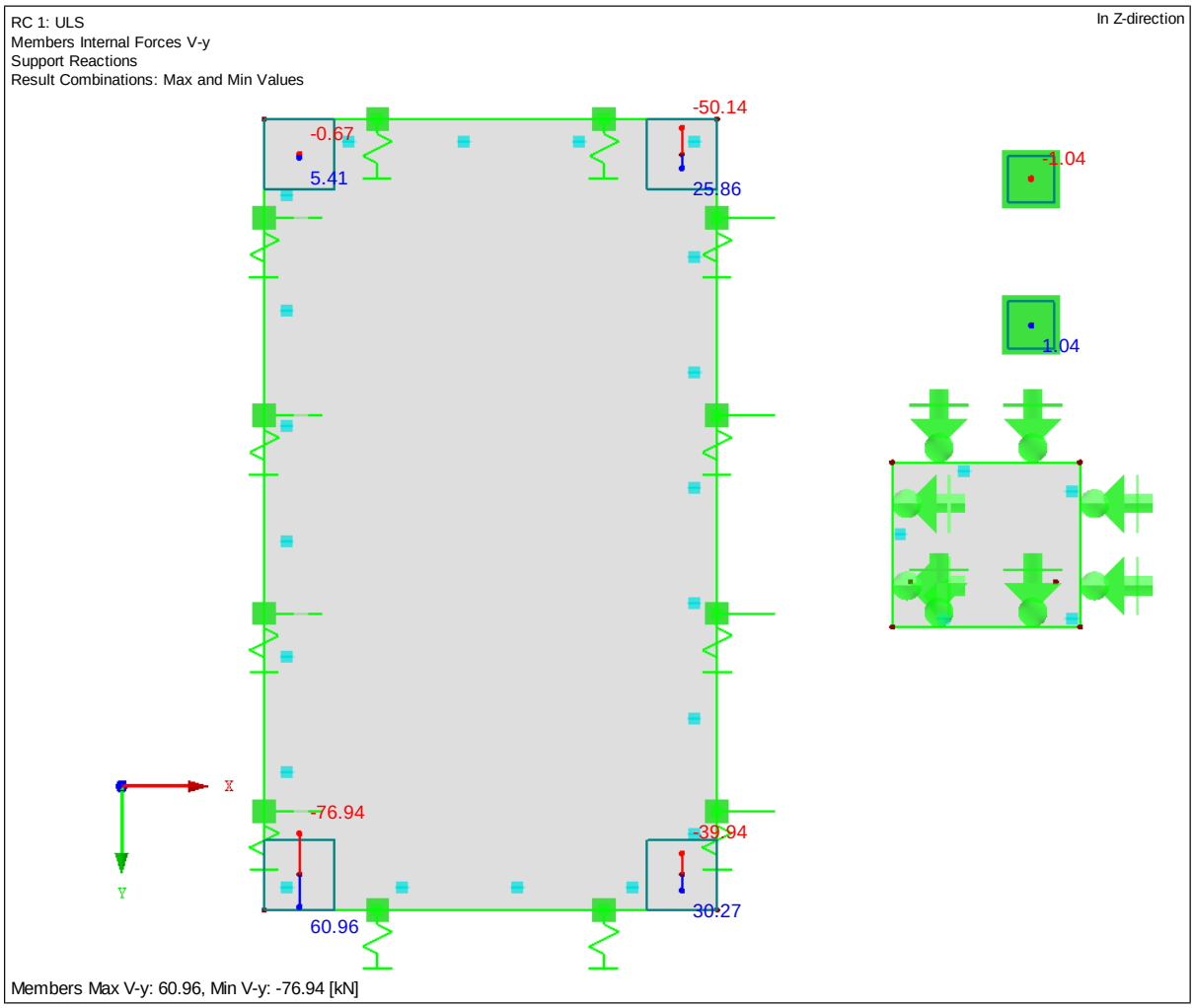


Figure 30 Shear force Vy,max

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

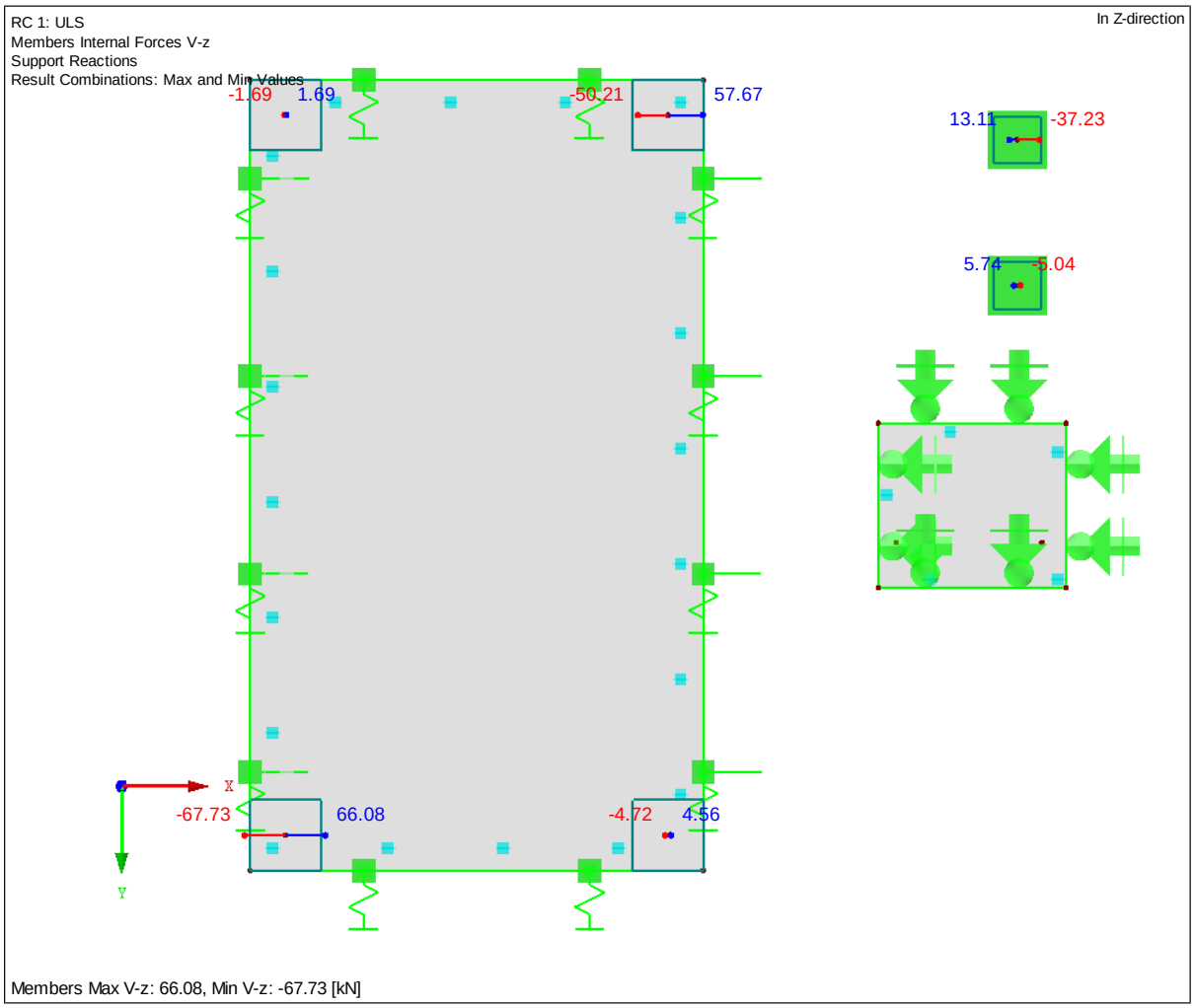


Figure 31 Shear force Vz,max

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

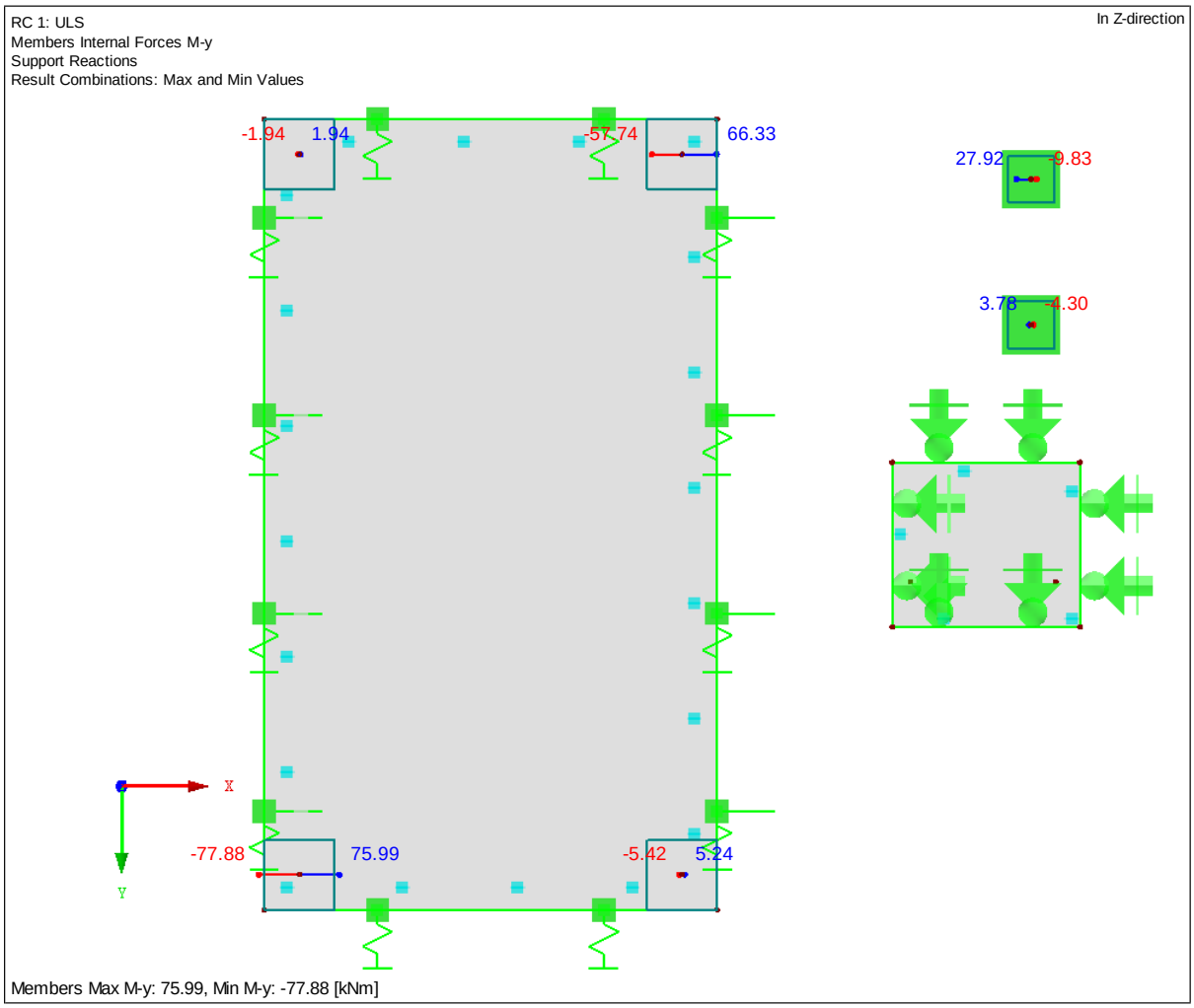


Figure 32 Bending moment My,max

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

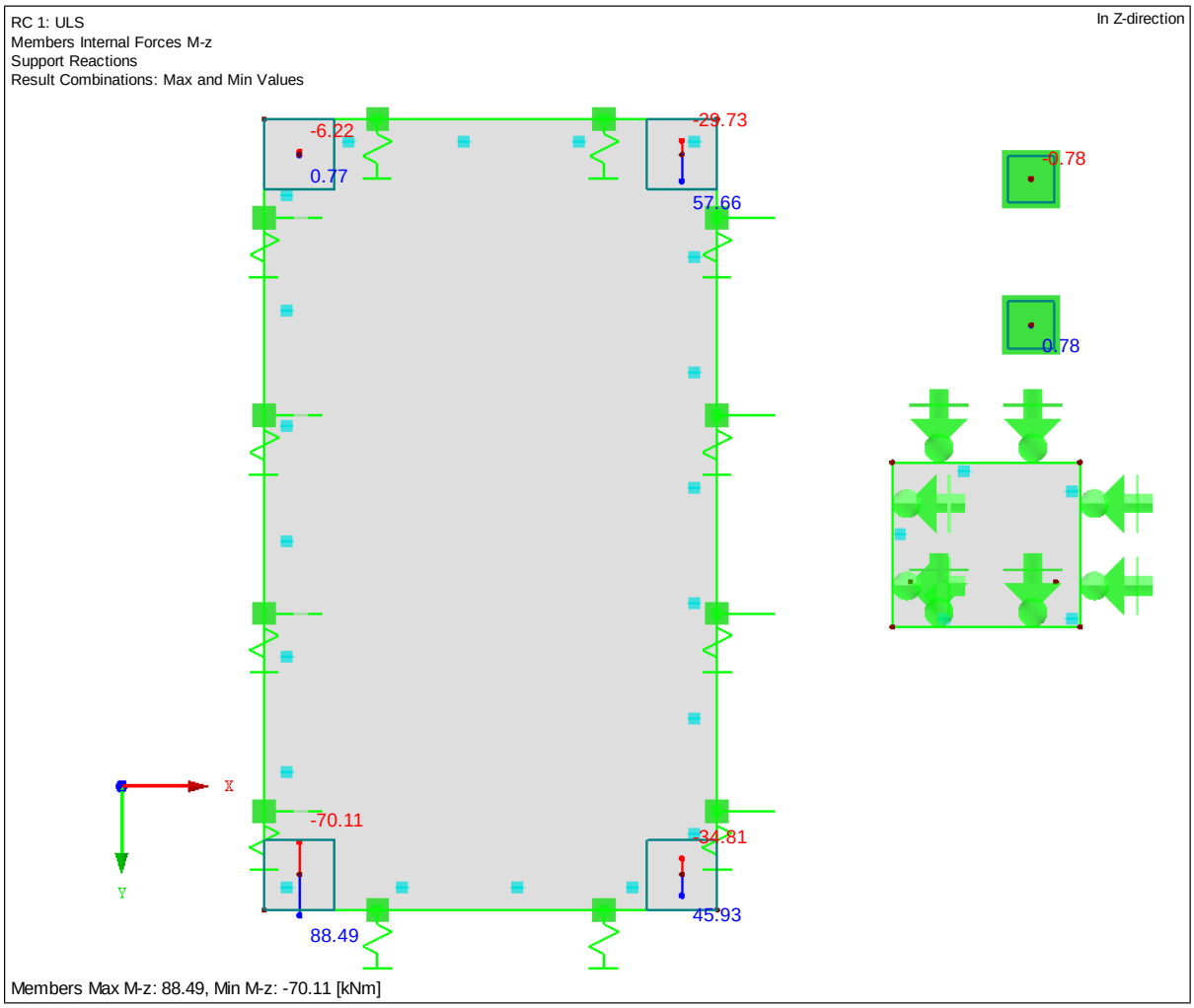


Figure 33 Bending moment Mz,max

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

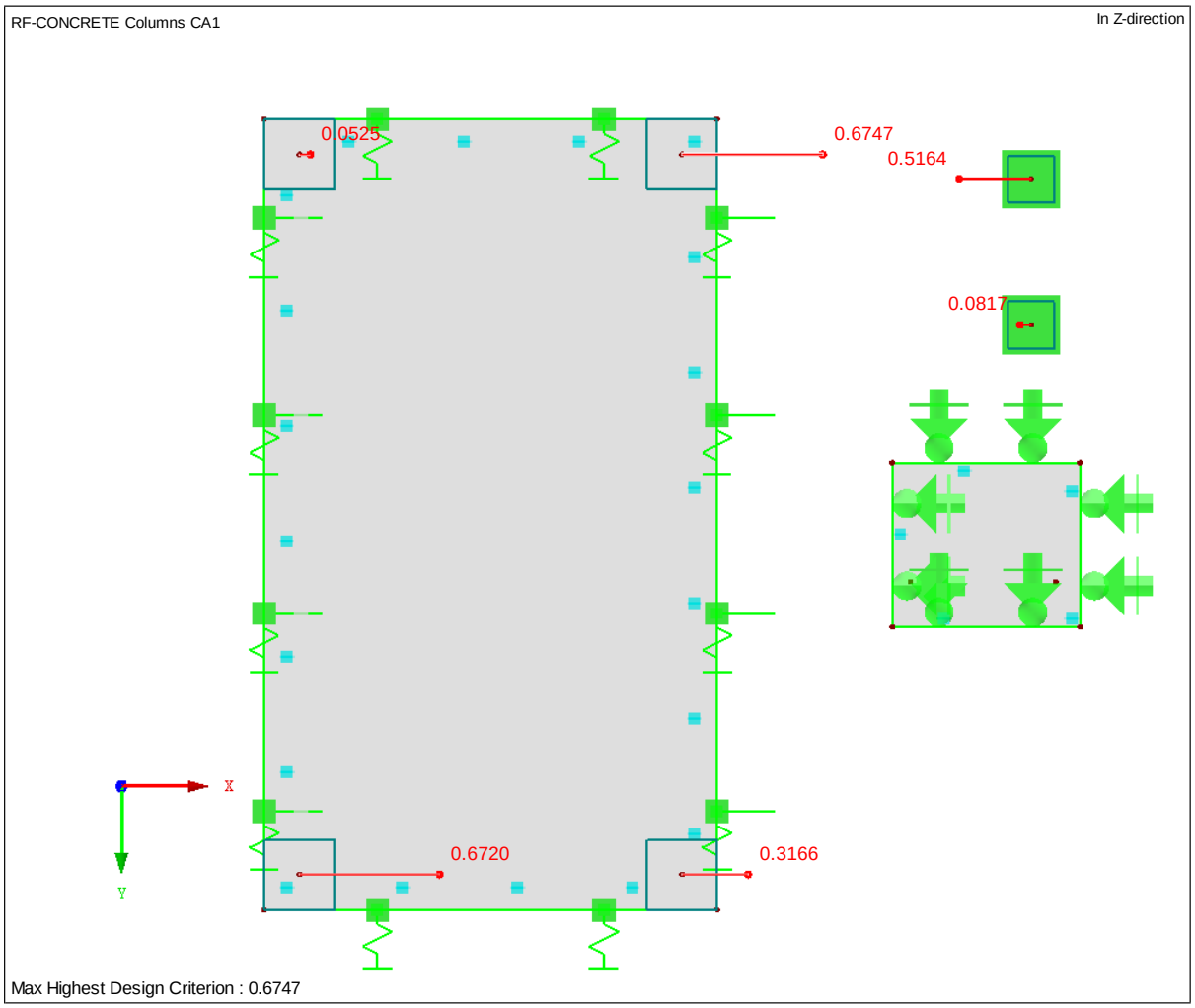


Figure 34 Highest utility ratio

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

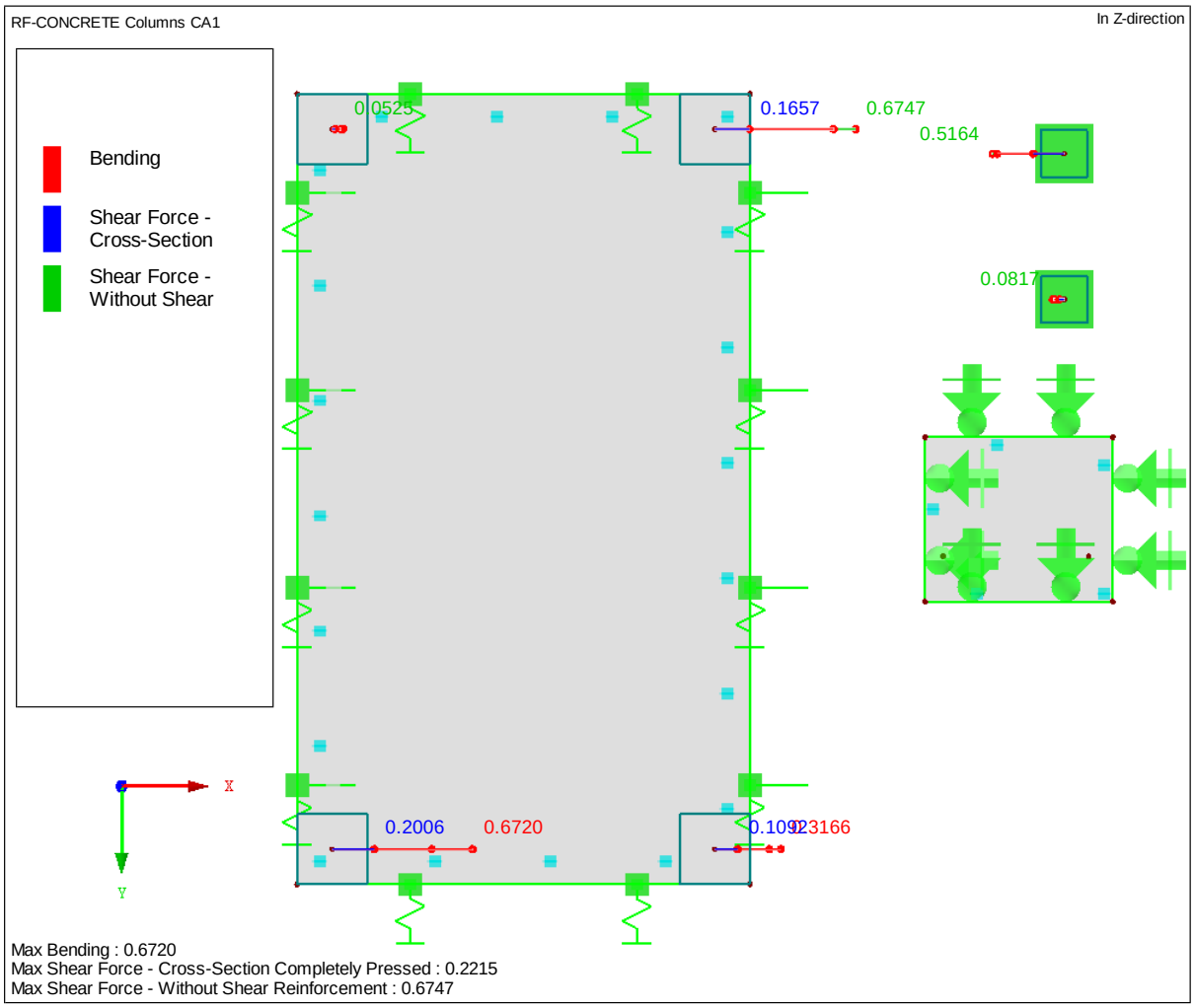


Figure 35 Utility ratio for bending and shear

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

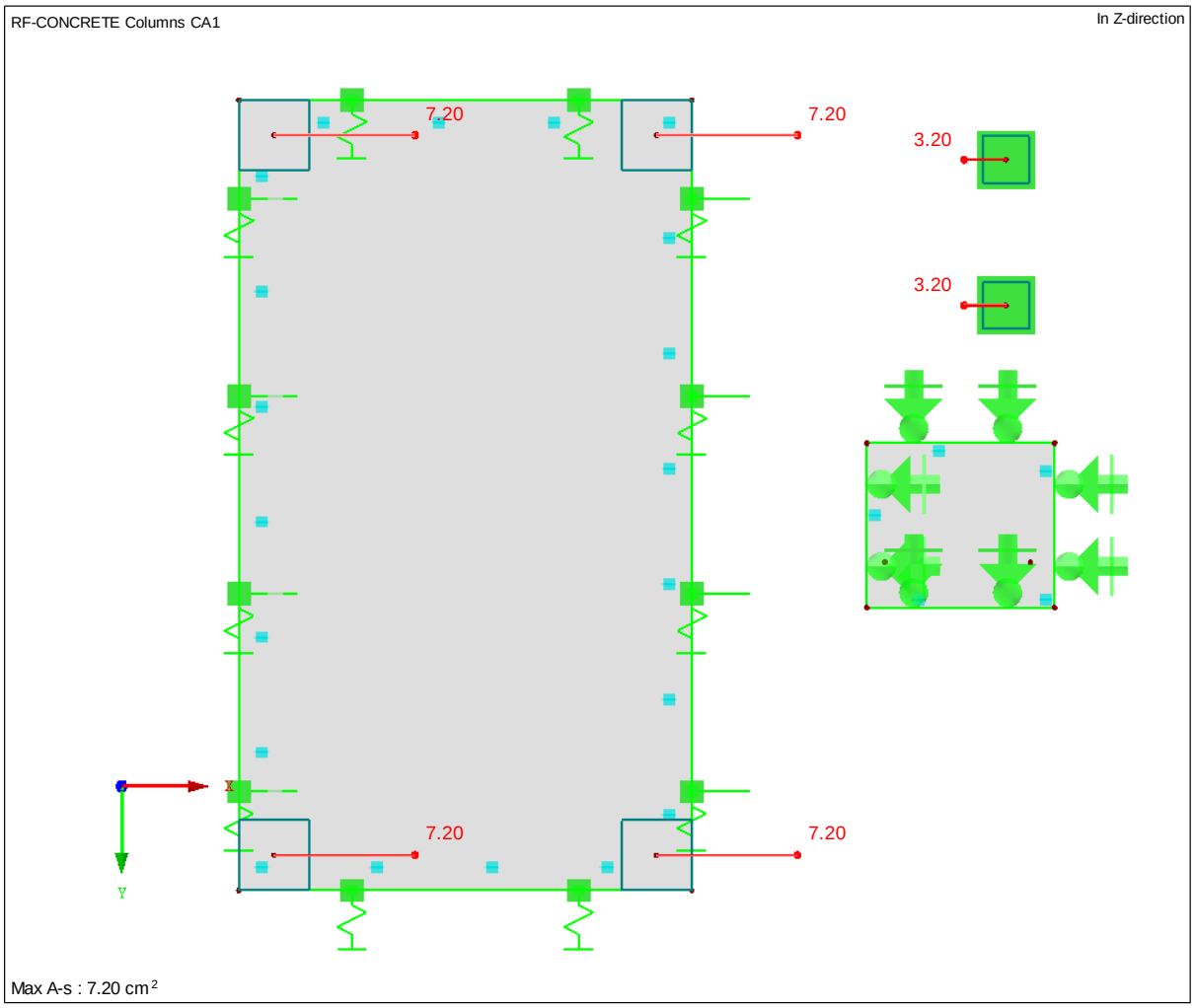


Figure 36 Required longitudinal reinforcement area As

Provide 8T16, Asp = 8* π* (16mm/2)^2 = 16.08cm^2 for combined footing

Provide 8T16, Asp = 8* π* (12mm/2)^2 = 9.05cm^2 for pedestal

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

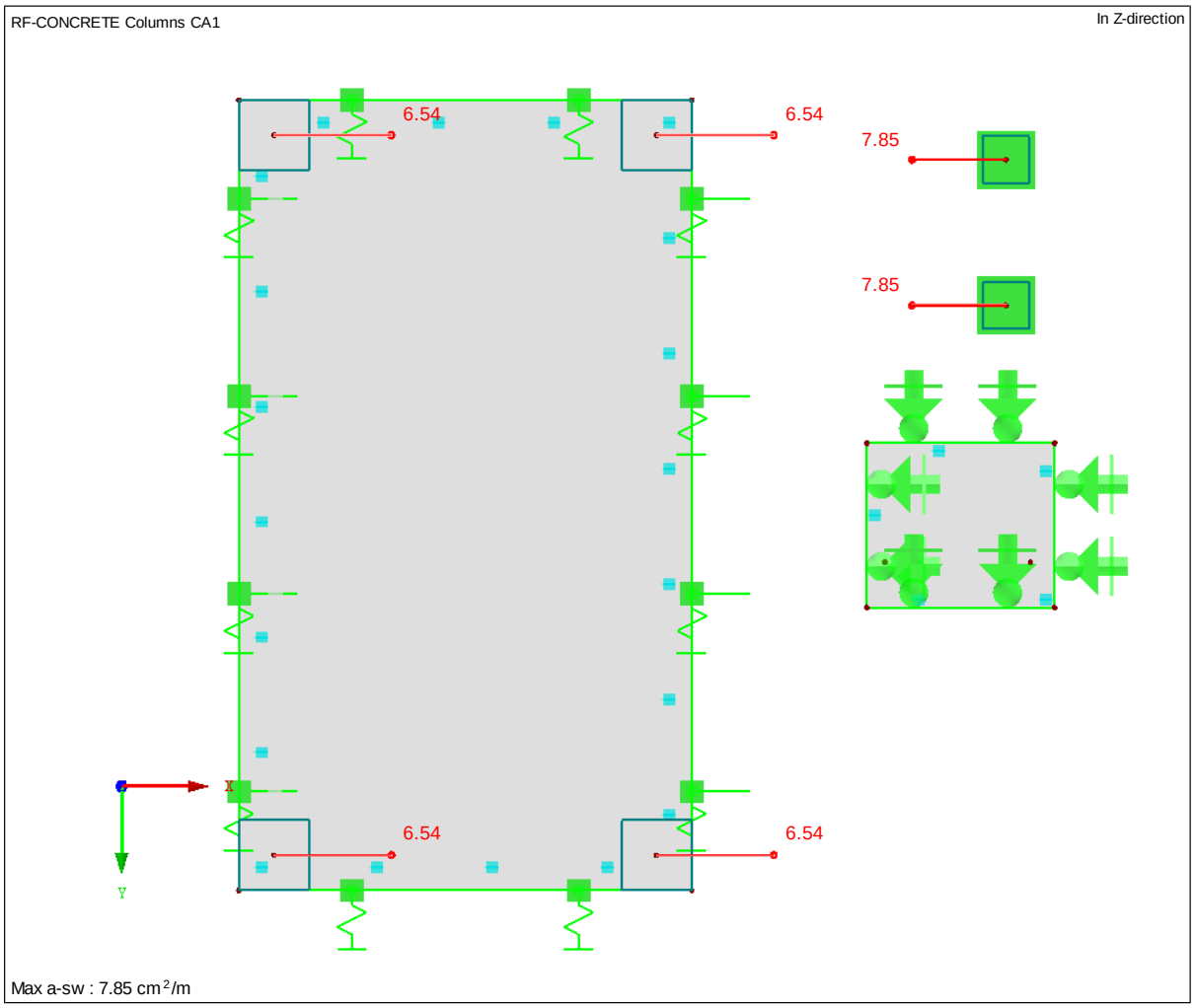


Figure 37 Required shear reinforcement area a_{sw}

Provide T12@150mm, As = 8* π* (12mm/2)^2 = 9.05cm^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6.6. Design of pad footing

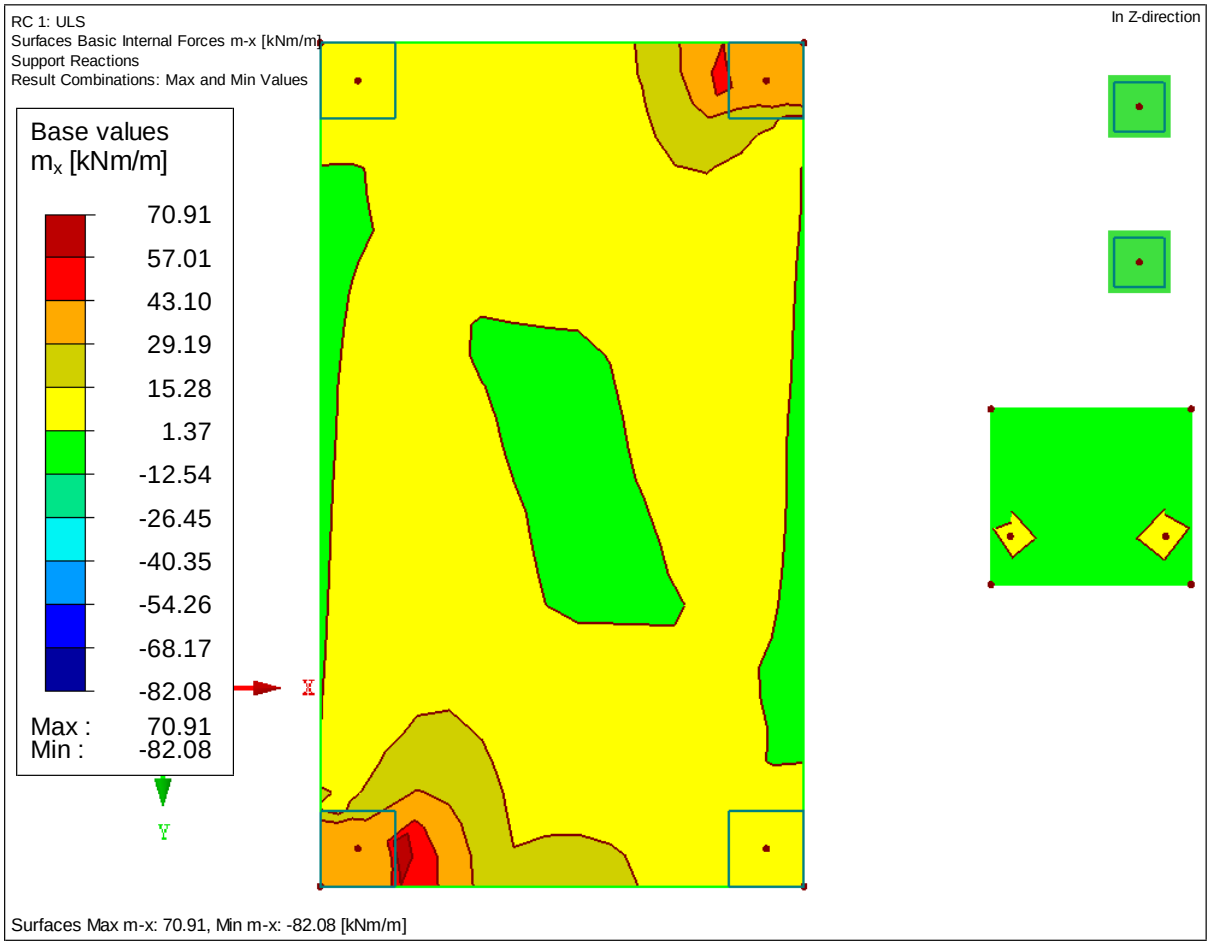


Figure 38 max bending moment Mx

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

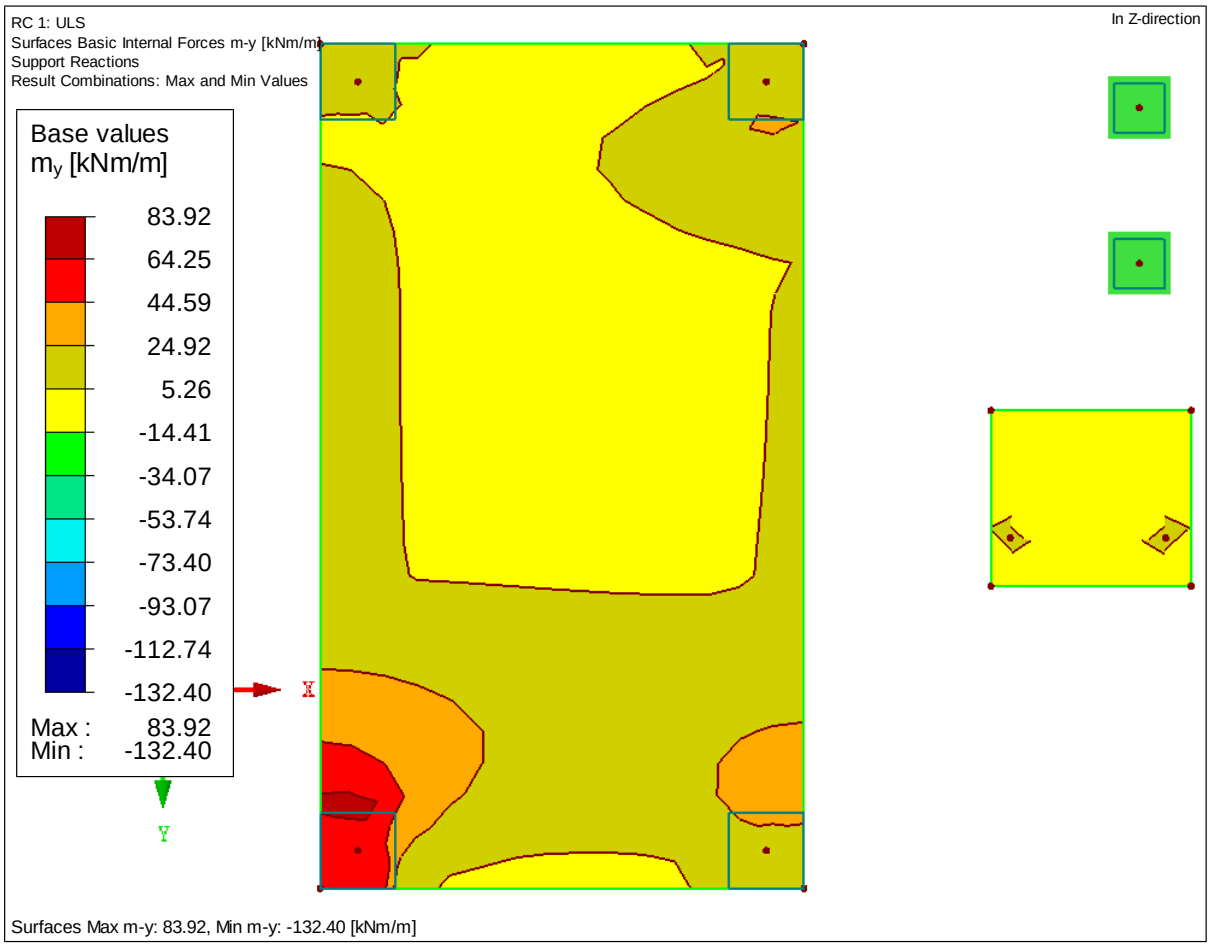
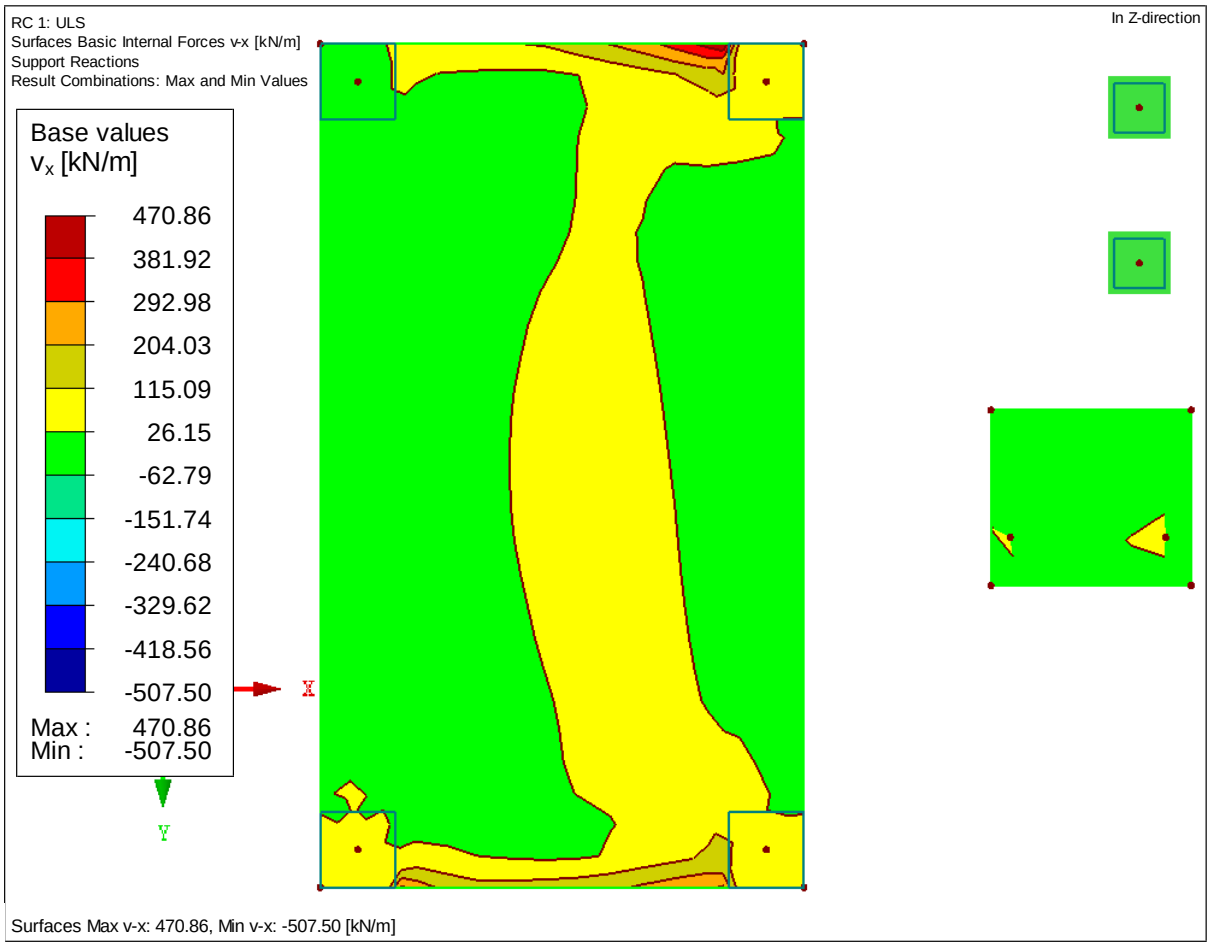


Figure 39 max bending moment My

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



C01 *Figure 40 Shear force Vx*

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

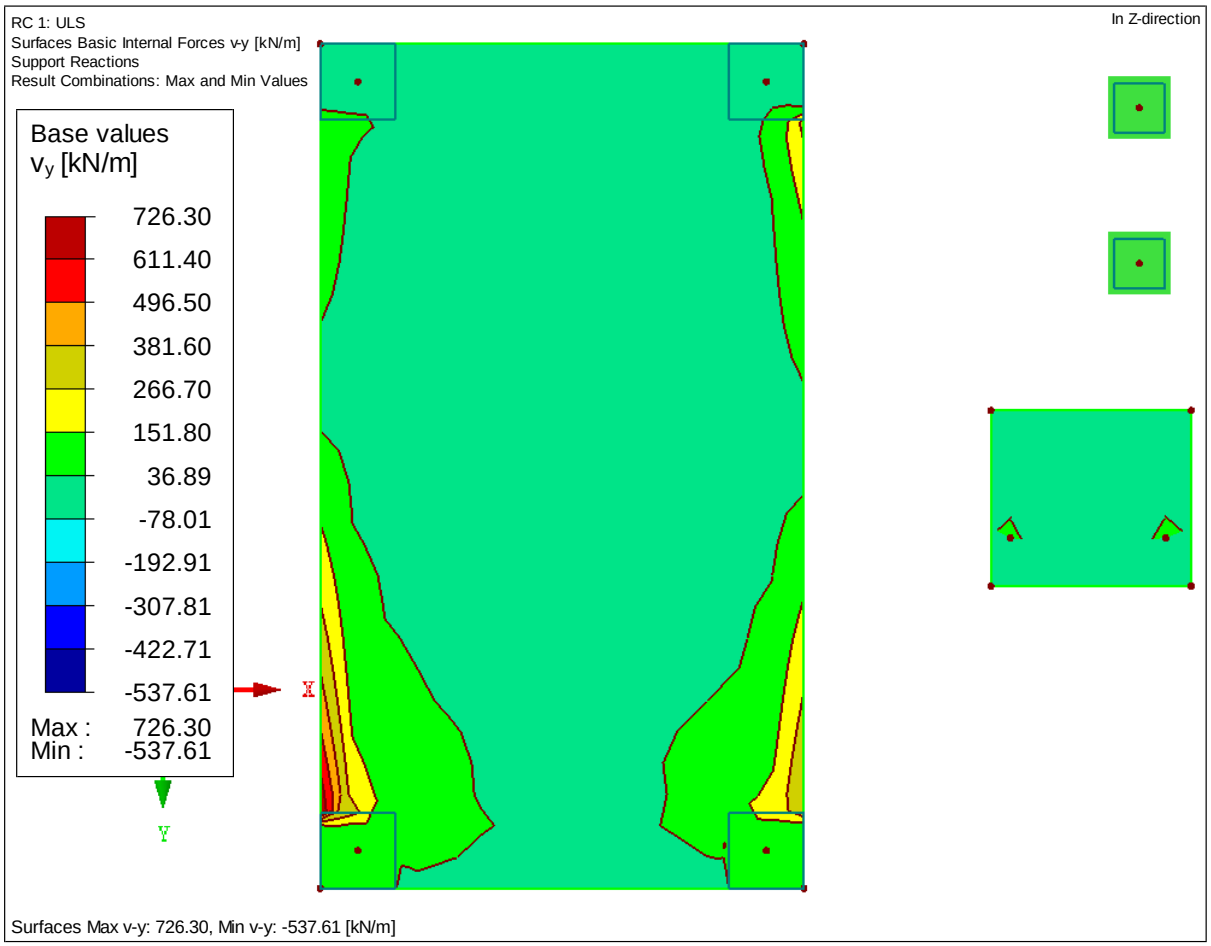


Figure 41 Shear force Vy

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

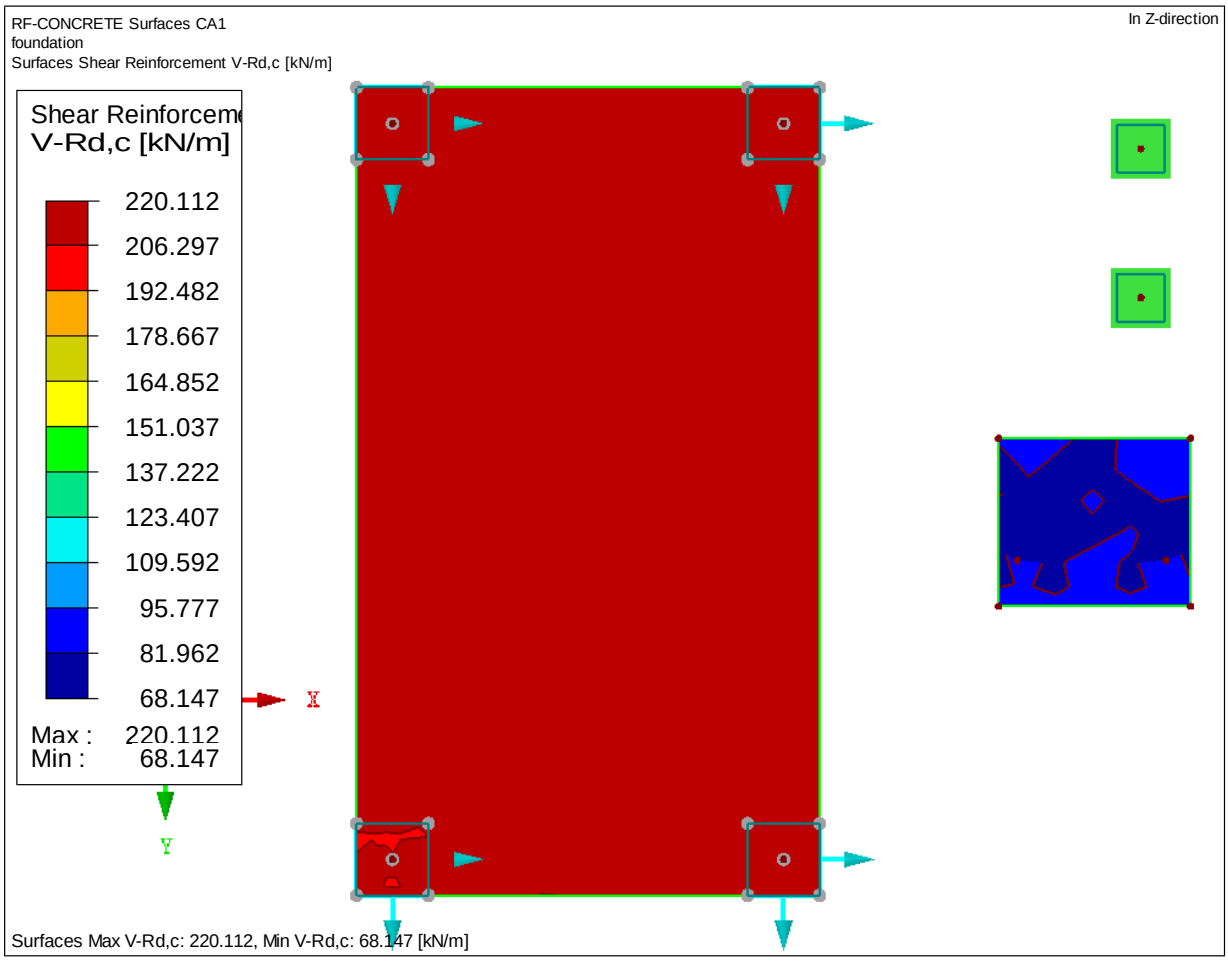


Figure 42 Shear resistance without shear reinforcement $V_{Rd,c}$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

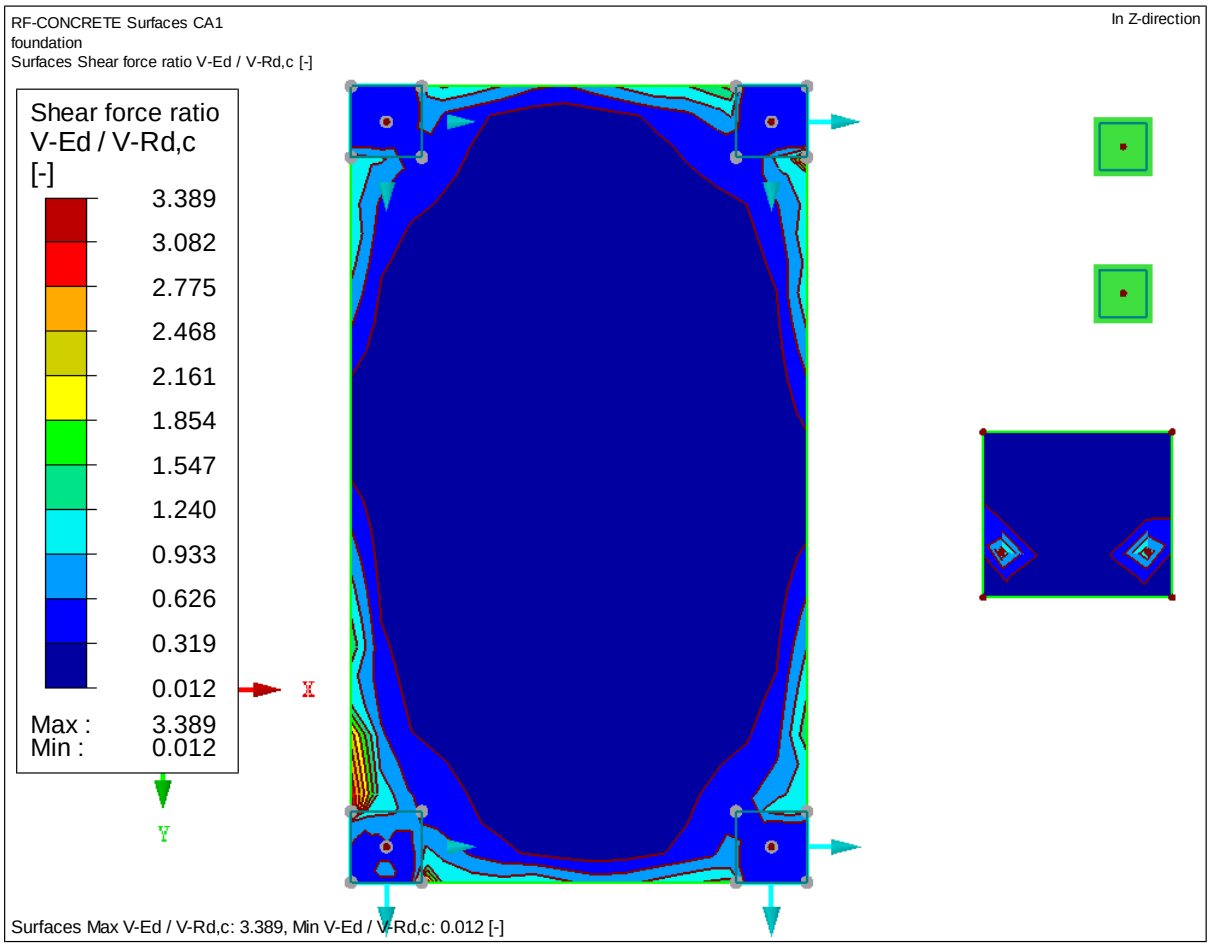


Figure 43 Utility ratio $V_{Ed}/V_{Rd,c}$

Utility ratio $V_{Ed}/V_{Rd,max}$ = applied shear force/ design shear resistance without shear reinforcement >1.0, shear reinforcement is required

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

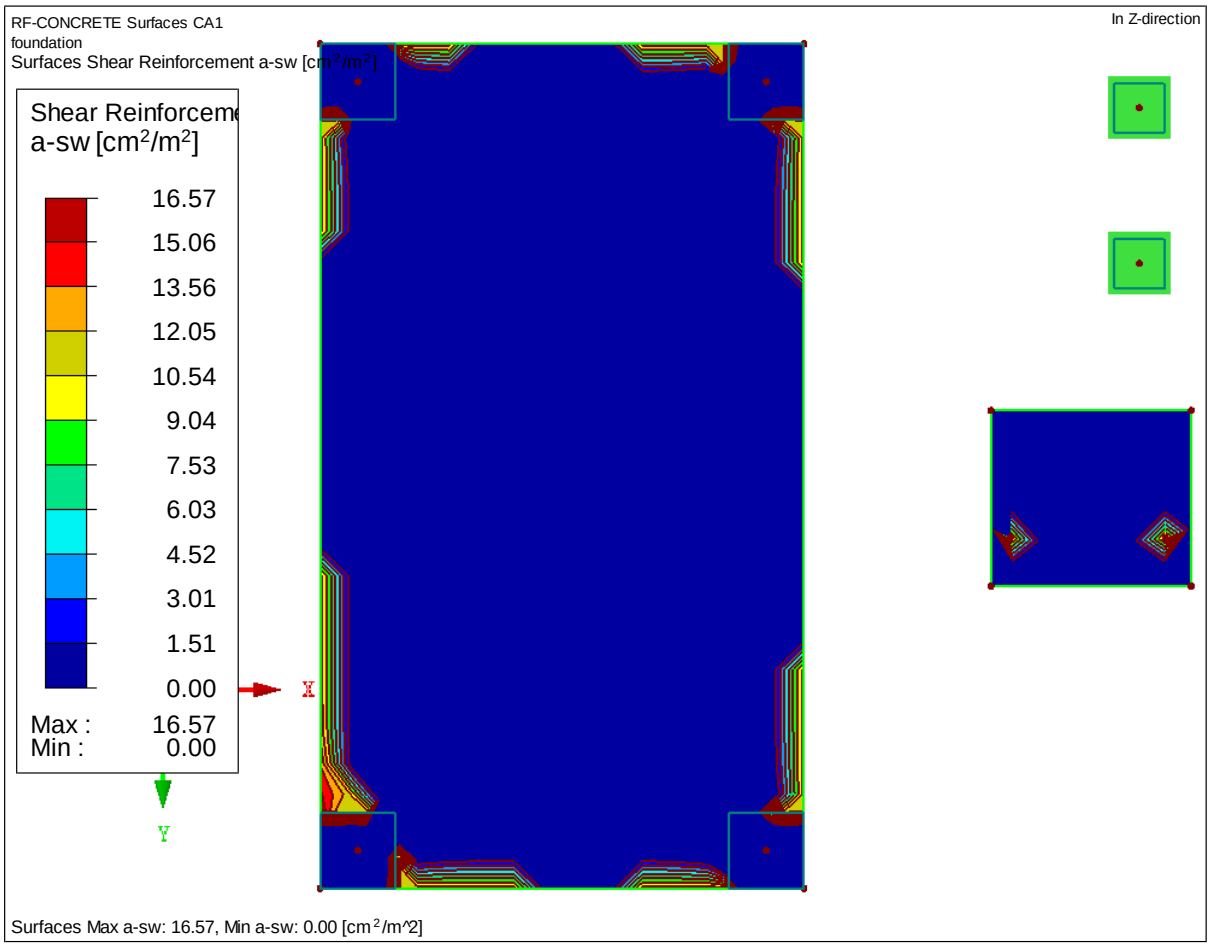


Figure 44 Required shear reinforcement a_{sw}

The local peak is generated in calculation model due to usage of averaged region. However, the result values are much less in reality.

Provide T16@150mm as shear reinforcement, a_{sp} = 13,04cm^2/m. Starter bars in pedestals will increase shear capacity locally too.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

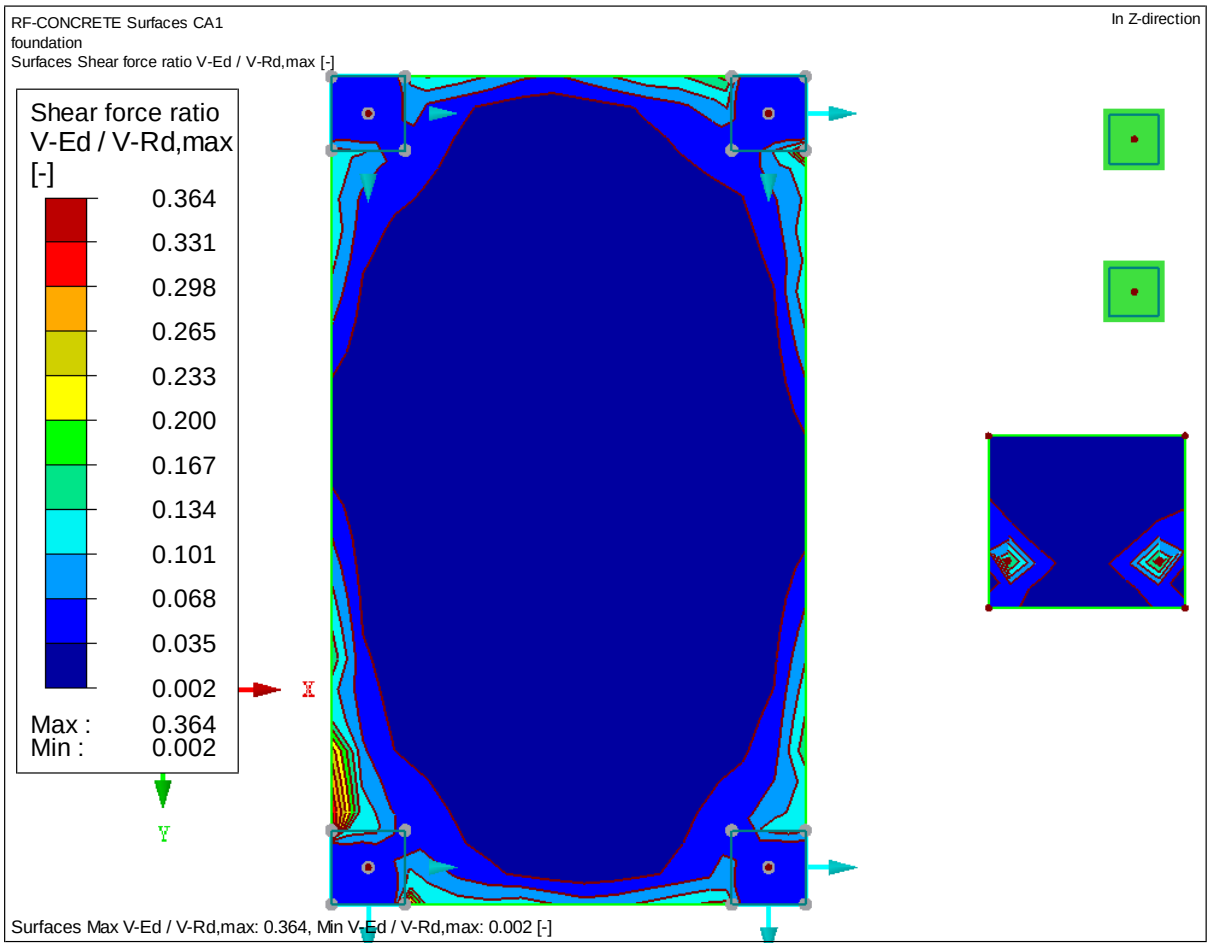


Figure 45 Utility ratio $V_{Ed}/V_{Rd,max}$

Utility ratio $V_{Ed}/V_{Rd,max}$ = applied shear force/ Design shear resistance of concrete compression strut <1.0, ok

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

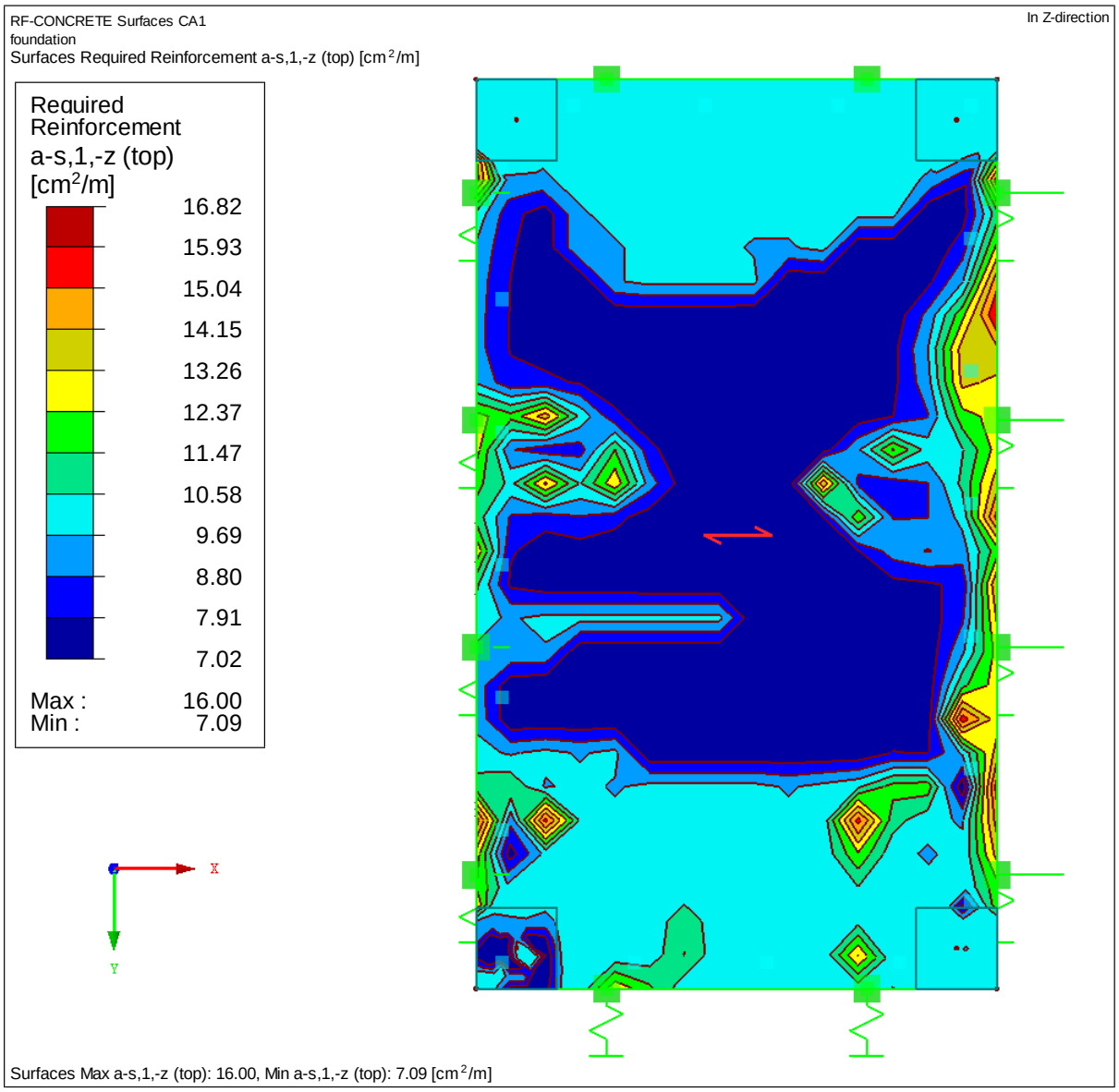


Figure 46 Required top reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

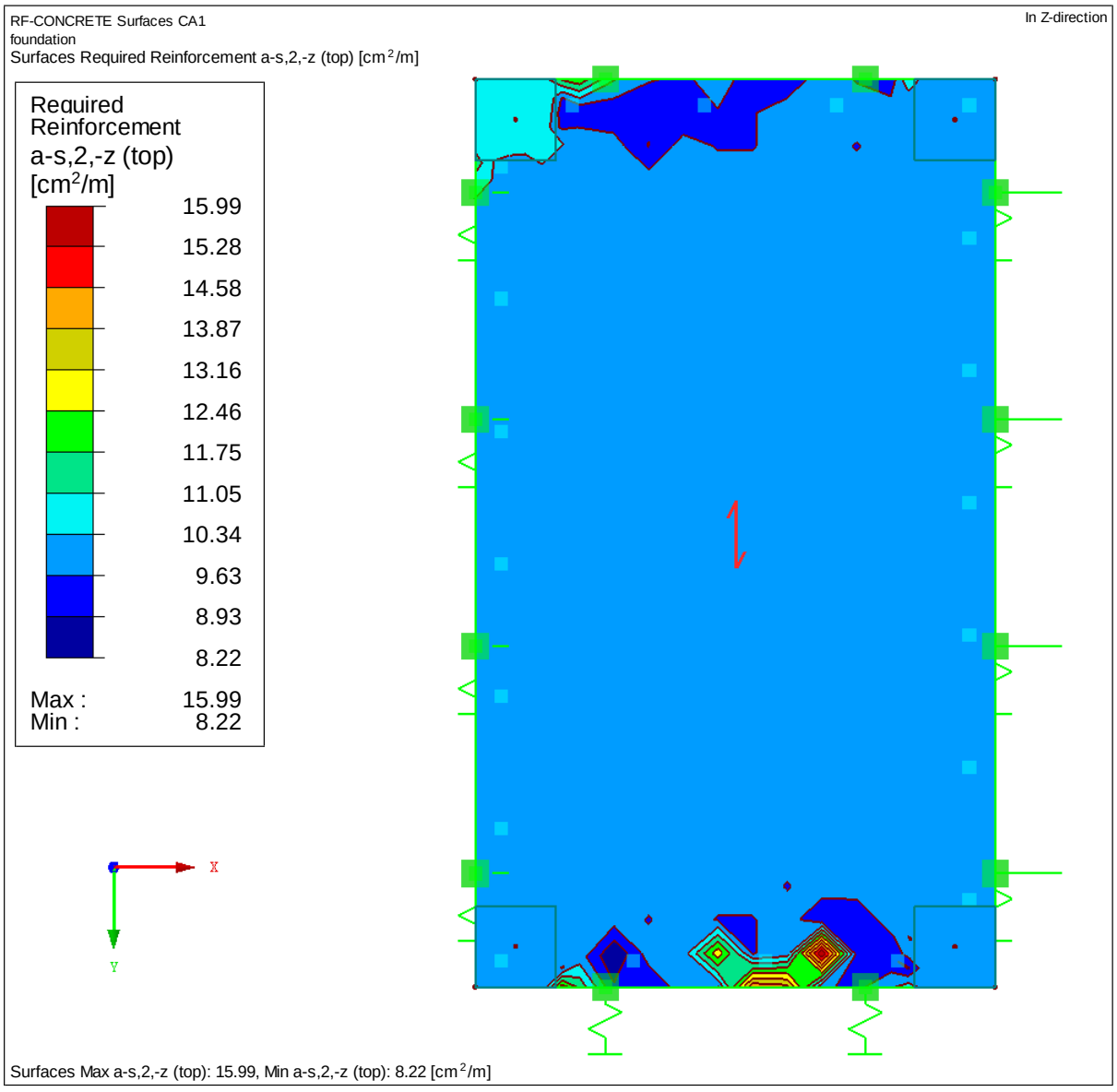


Figure 47 Required top reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

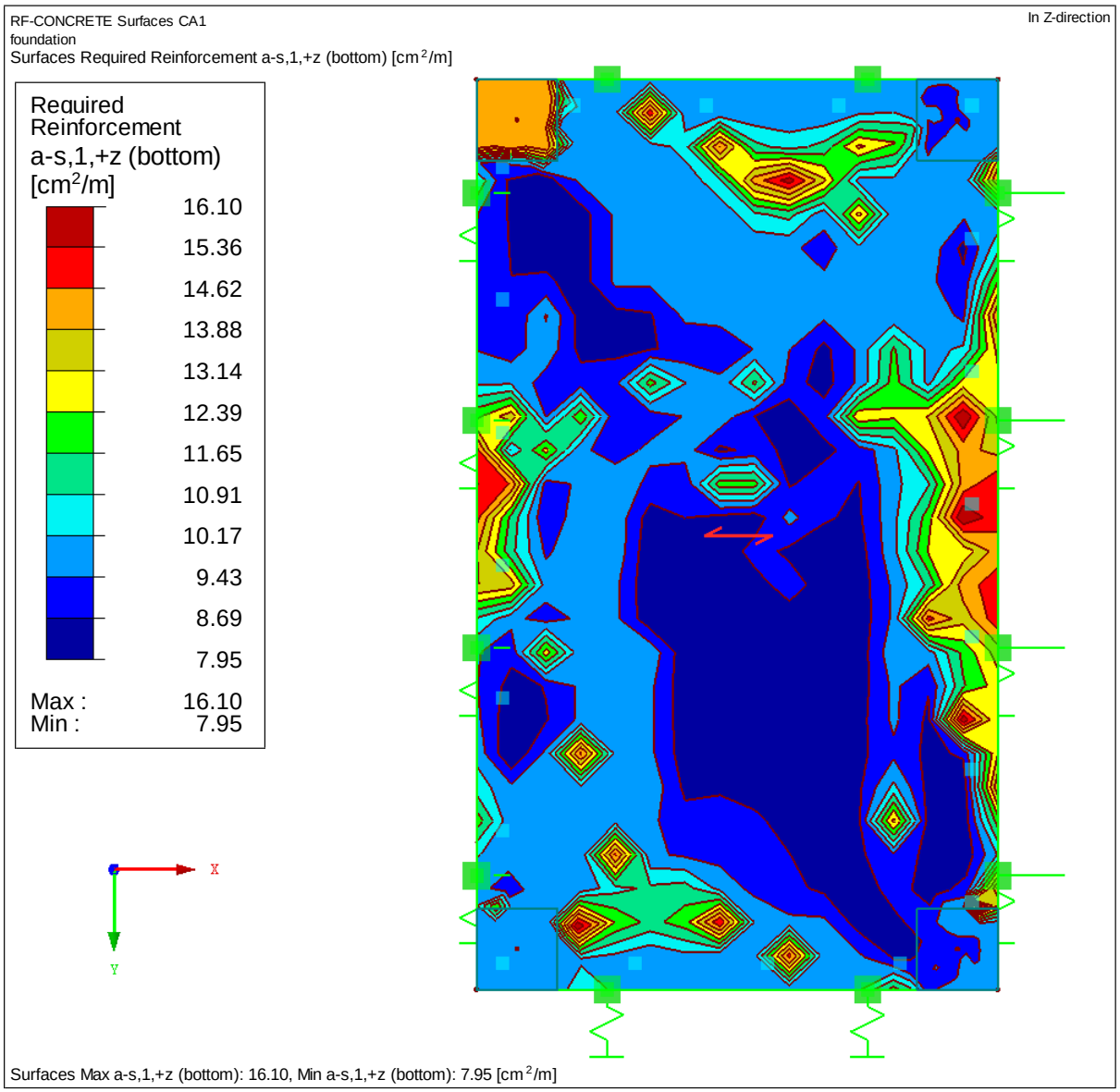


Figure 48 Required bottom reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

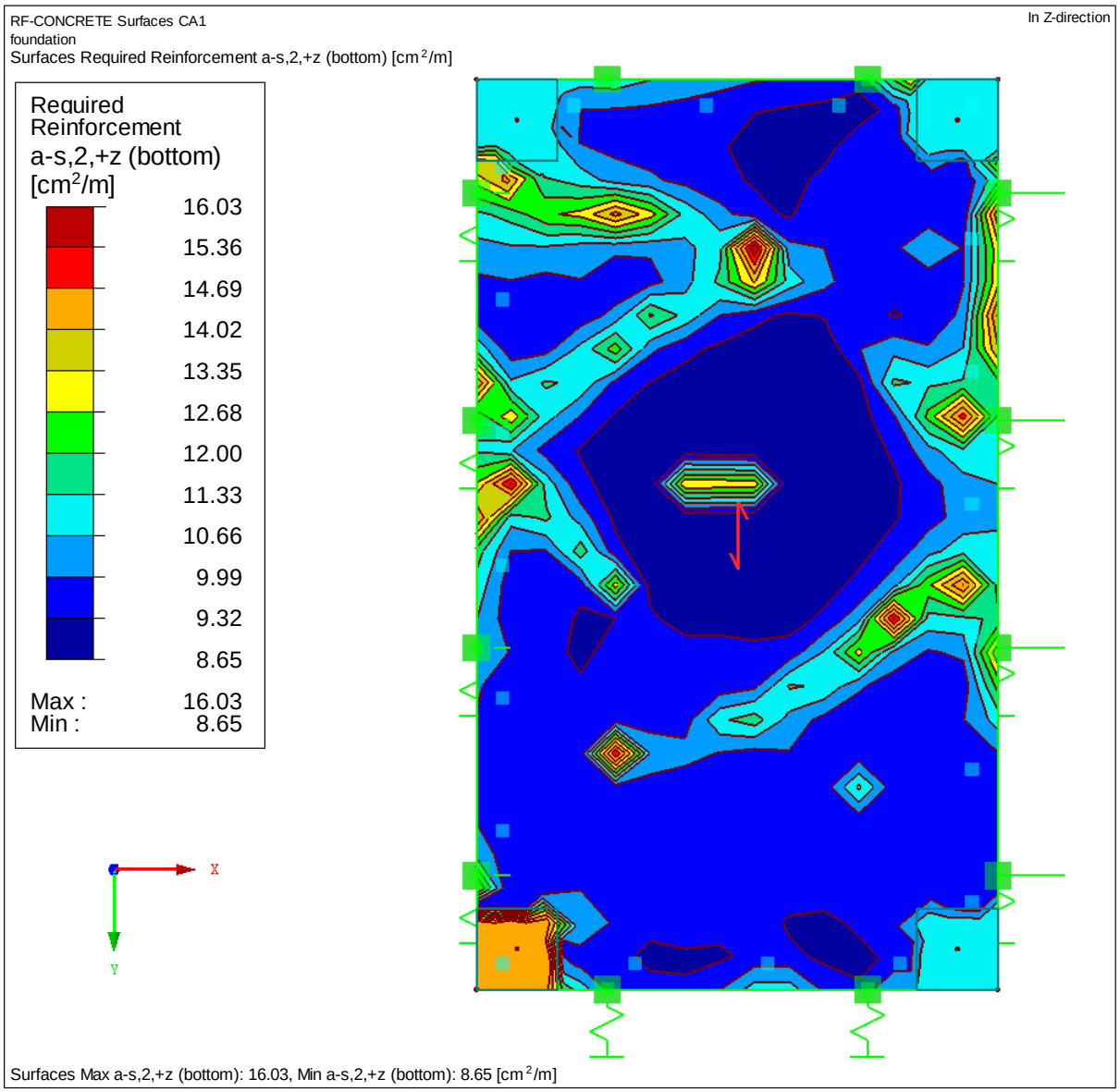
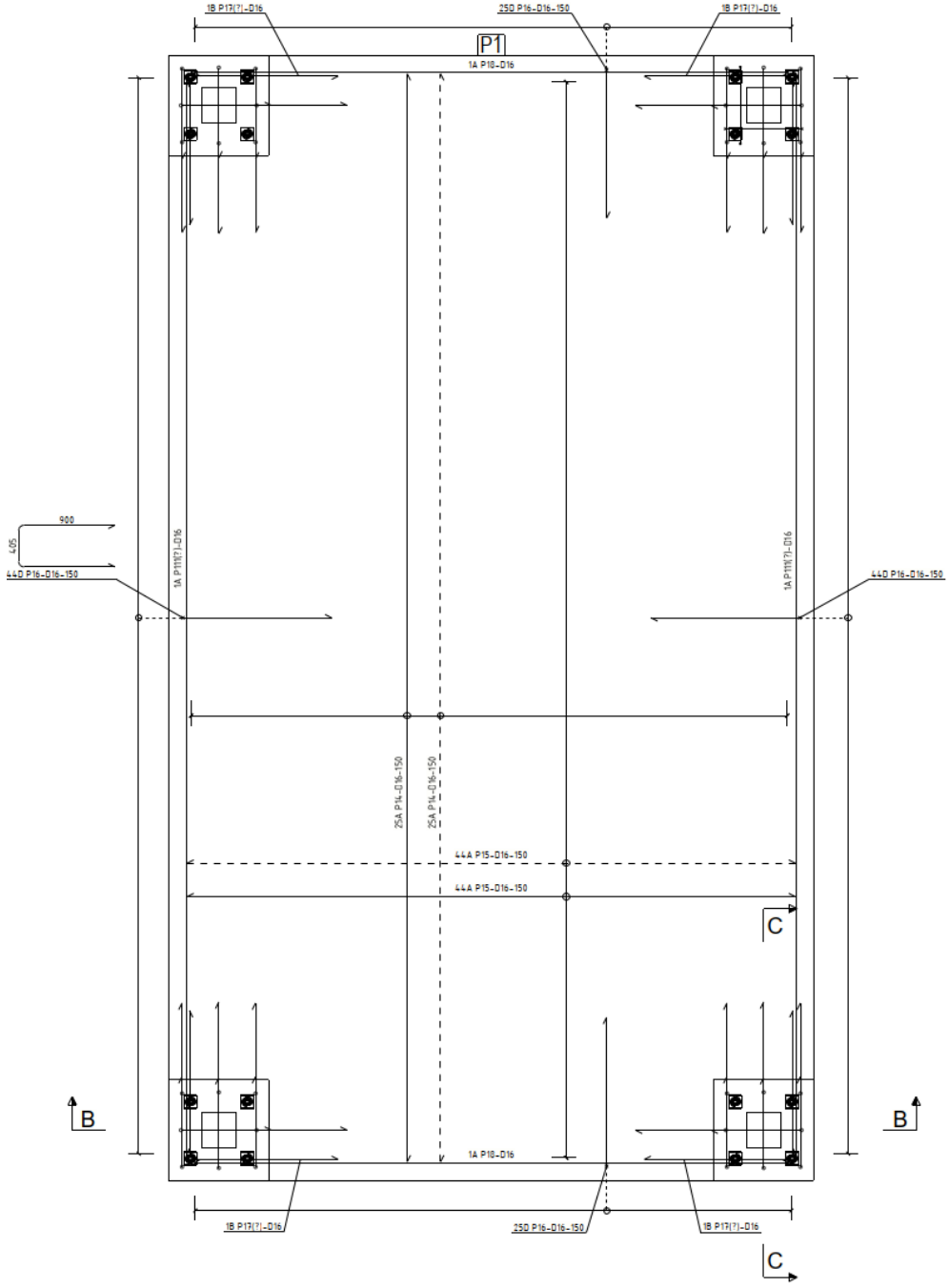


Figure 49 Required bottom reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



C01 *Figure 50 Sketch of foundation reinforcement (top)*

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

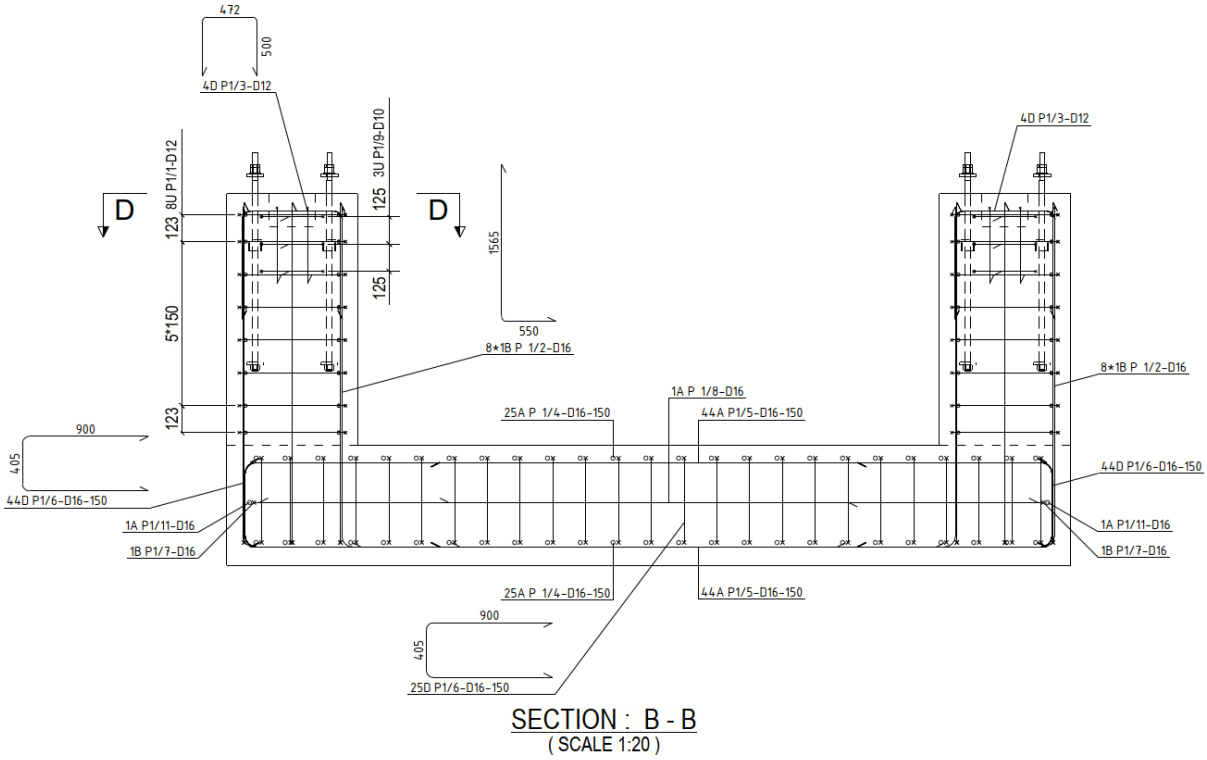


Figure 51 Sketch of foundation reinforcement (section)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

[1] Saudi Building Code (SBC) 301, Structural - Loading and Forces.

[2] Saudi Building Code (SBC) 304, Structural – Concrete Structures.

[3] Project Design Criteria, MD-216-1000-EG-ST-CRT-1001-A03.

[4] Overall geotechnical report, MD-216-3000-EG-CV-RPT-1008-C01.

[5] General arrangement drawing, raw water treatment plan, 1-8/E-J, MD-216-49W1-EG-CV-GAD-1001

[6] Detail drawing, raw water treatment plan, staircase foundations, MD-216-49W1-EG-CV-DED-1006

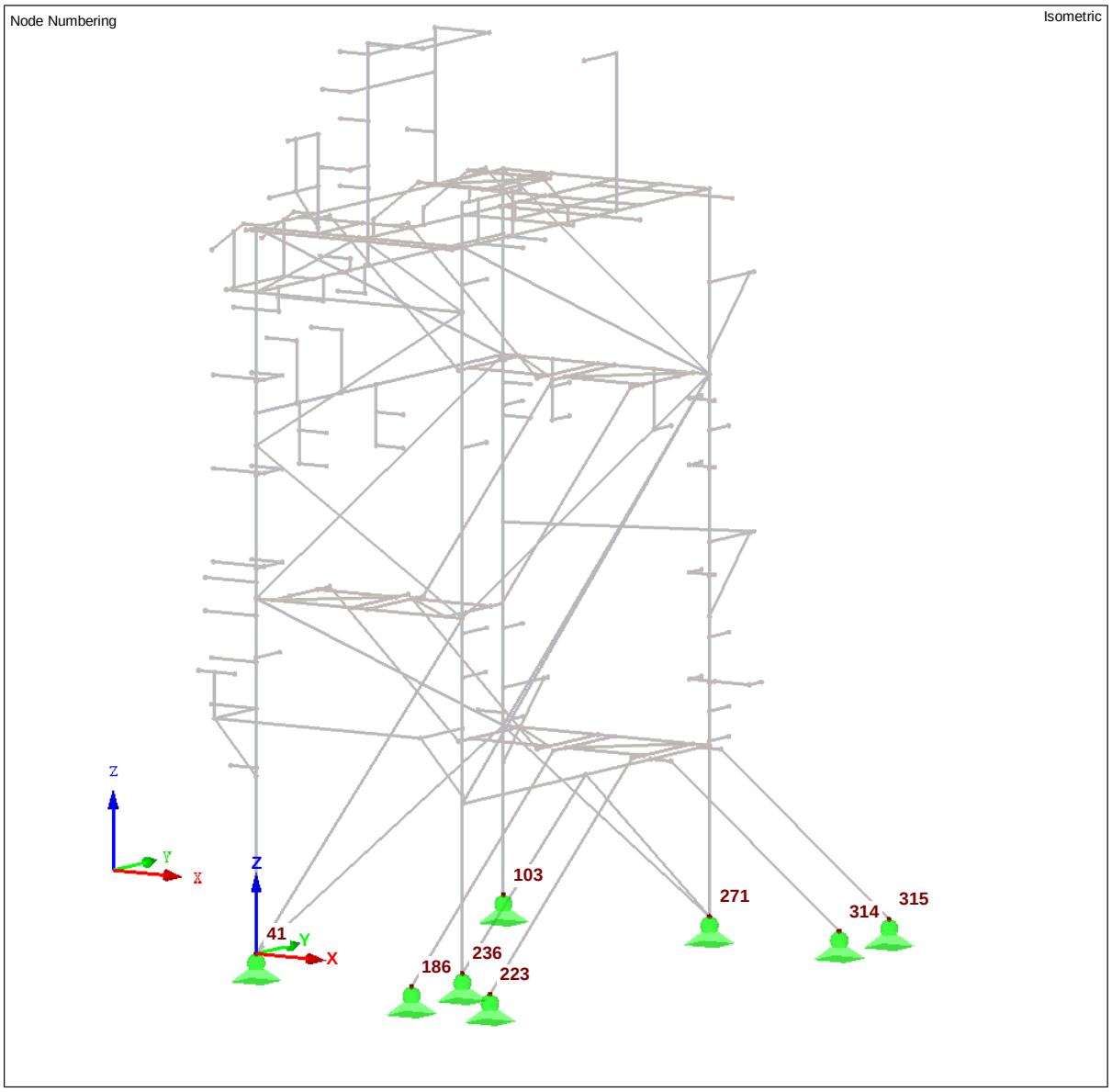
[7] General arrangement drawing, raw water treatment plant, buffer tank's staircase & sump pump, MD-216-49W1-EG-ST-GAD-1001

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 INPUT LOAD DATA FROM STAIRCASE



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 2 COMMENT RESOLUTION SHEET

When comments are received from Employer, they need to be placed in a comment resolution table, and the filled table is always attached to the end of the document.

COMMENT TABLE	Document Name:	Document Number:	Revision No.	Comment Date:
General Comment:				

Sr. No	Chapter	Regarding	Comment	Resolution
1	4.1	live load	Please provide what the input values and assumptions are?	please find it in MD-216-49W1-EG-ST-CAL-1001
2	6.2	soil bearing pressure	Please draw a conclusion on bearing capacity check!	added
3	6.5	title	column to pedestal	changed
4	6.6	sketch	plain view	added

1 Remember also to always mark the made changes with a triangle to the left side of the document.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

You can also reply to the comments on an excel template, then it will be coded as a separate document and submitted to Employer together with the revised document itself. You can find the instruction and the excel template from “Document Codes and Numbering Procedure” [1].

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.